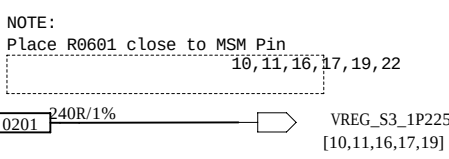
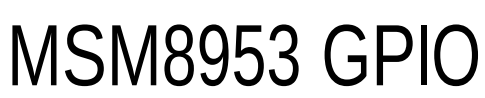


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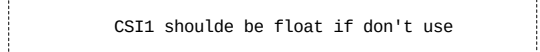
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MSM8953 EBI



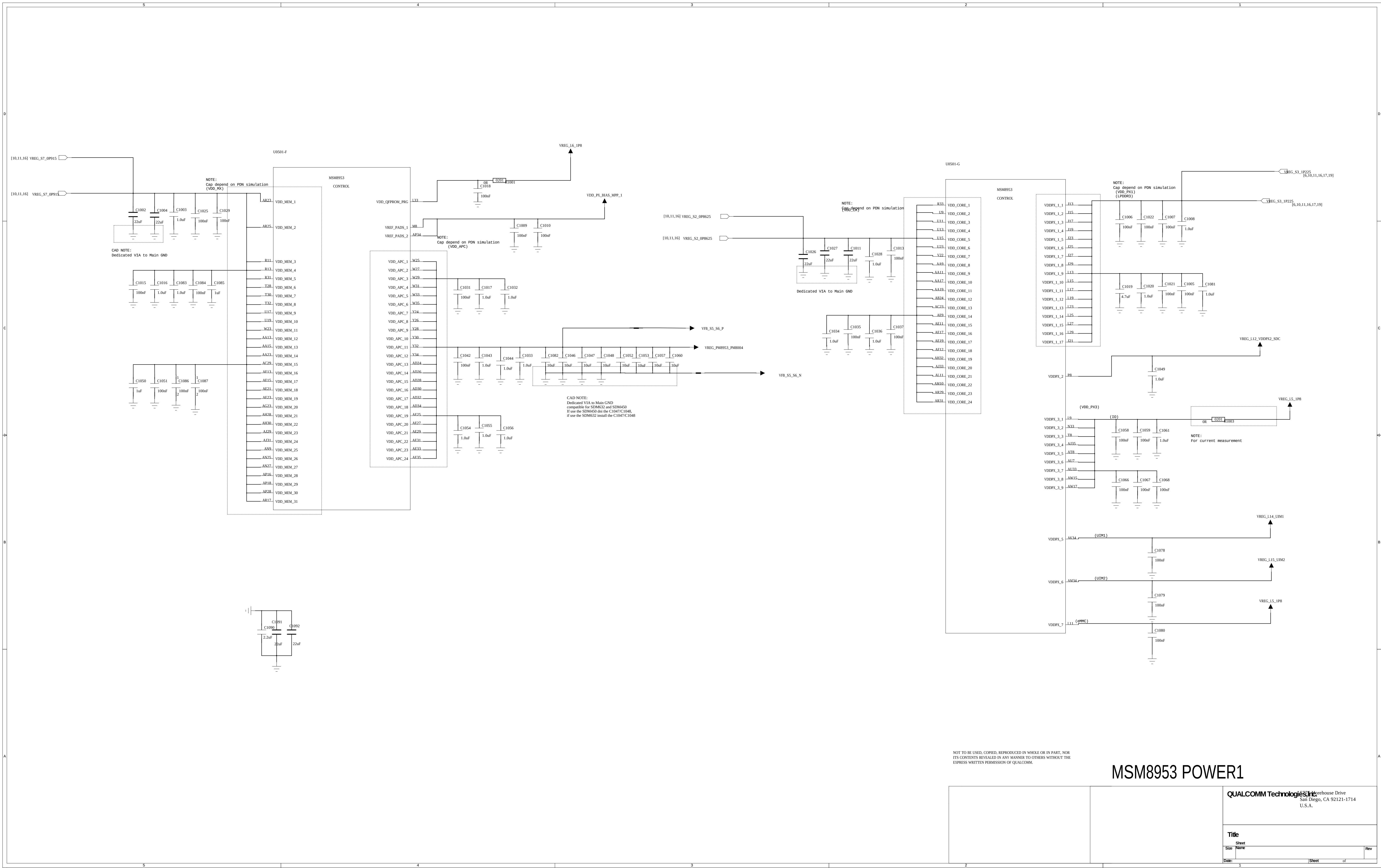
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MSM8953 POWER1

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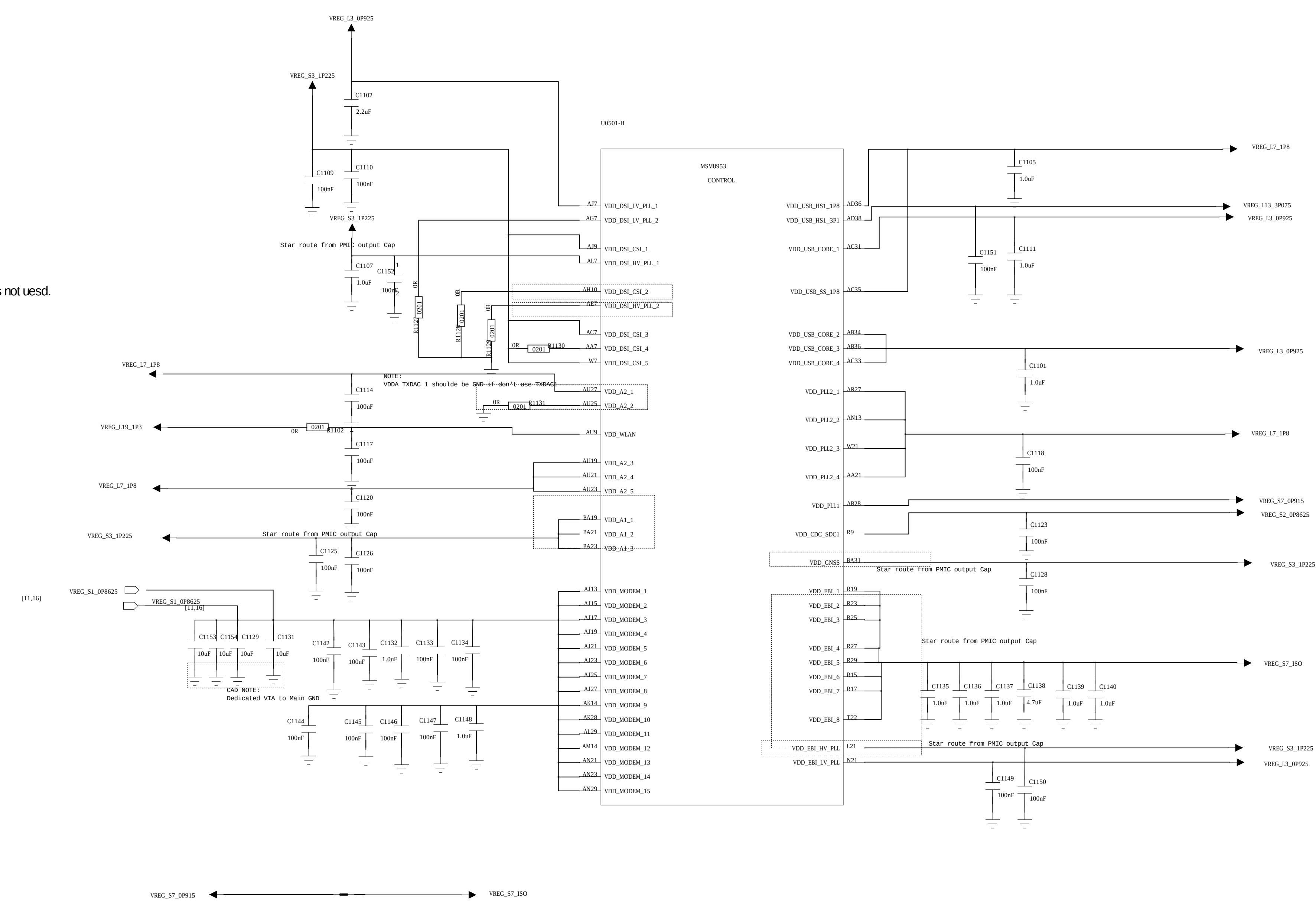
CAD NOTE:
About star route from PMIC output Cap:

Note: Connect SDM pins(AG7,AH10 and AE7) to GND if DS11 is not used.

Note: Respective Camera pins can be left open if not used.

Float AC7 pin if CAM0 not used

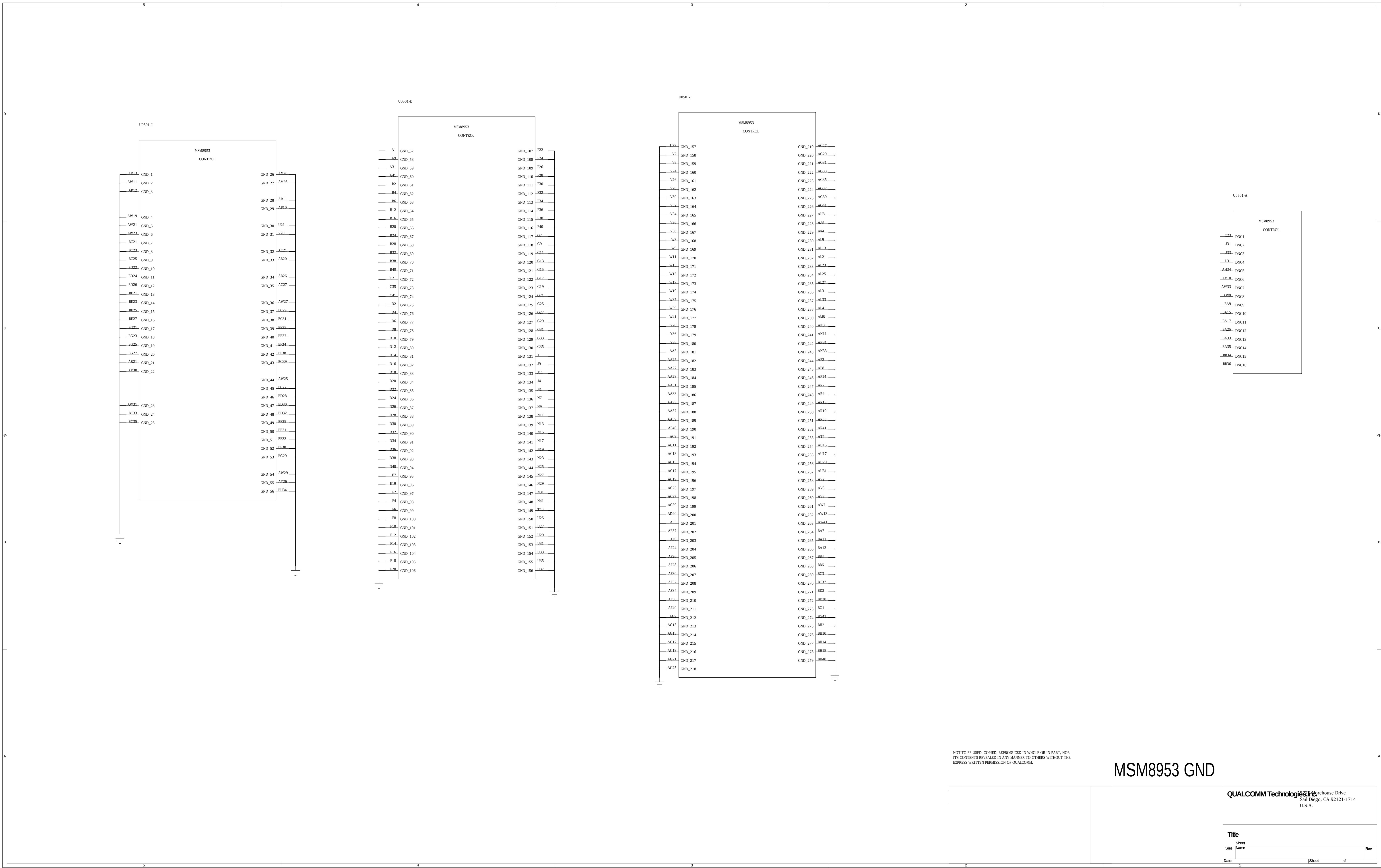
Float W7 pin if CAM2 not used



MSM8953 POWER2

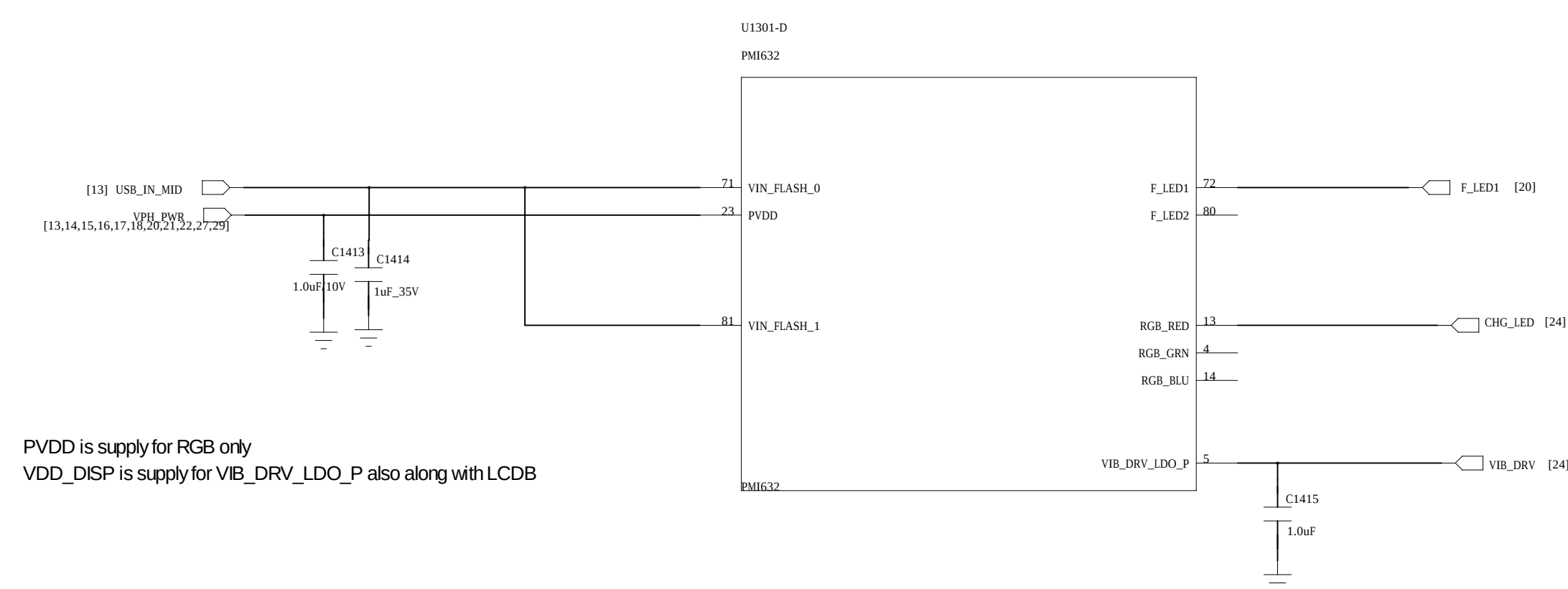
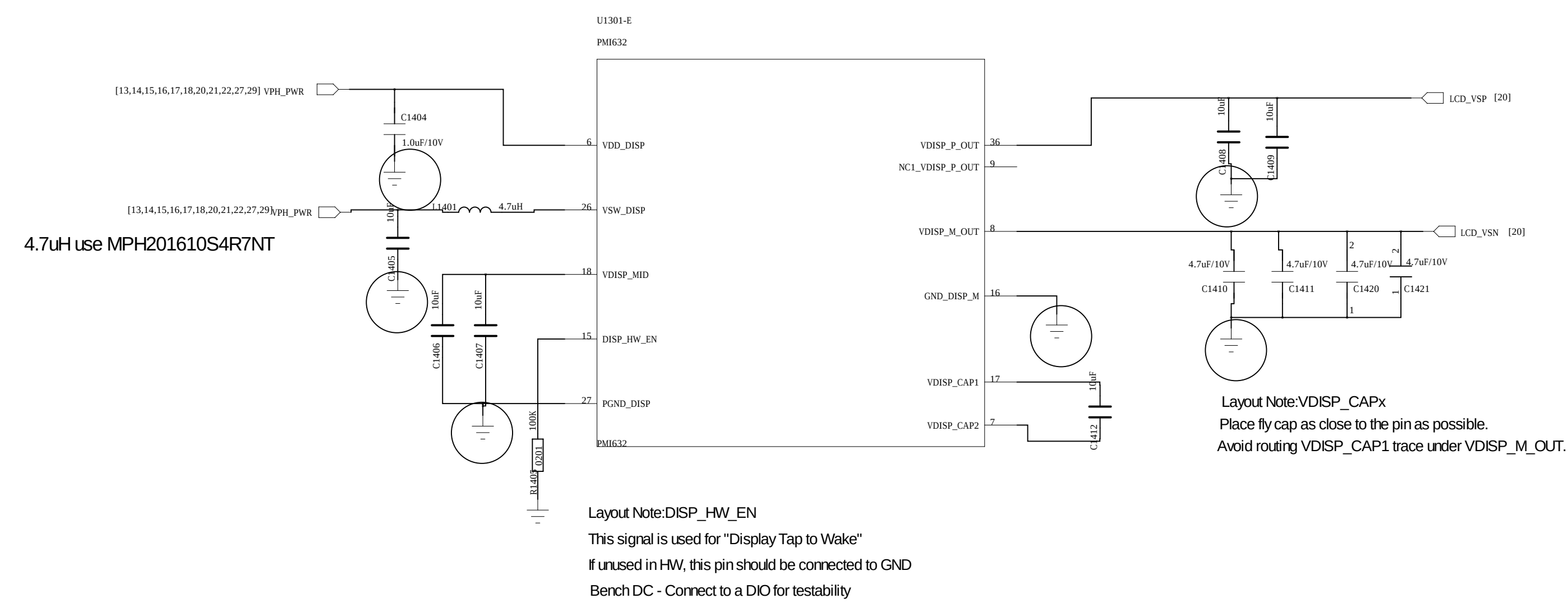
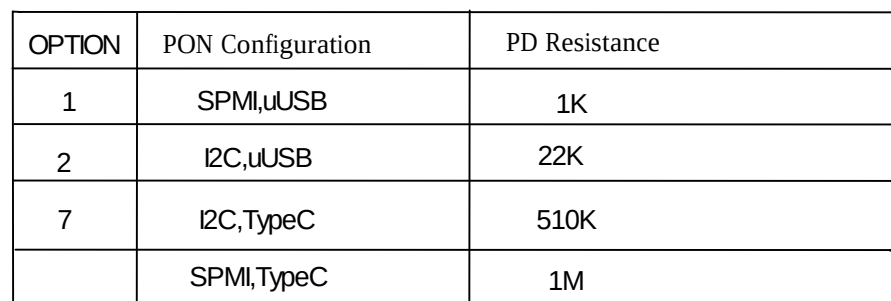
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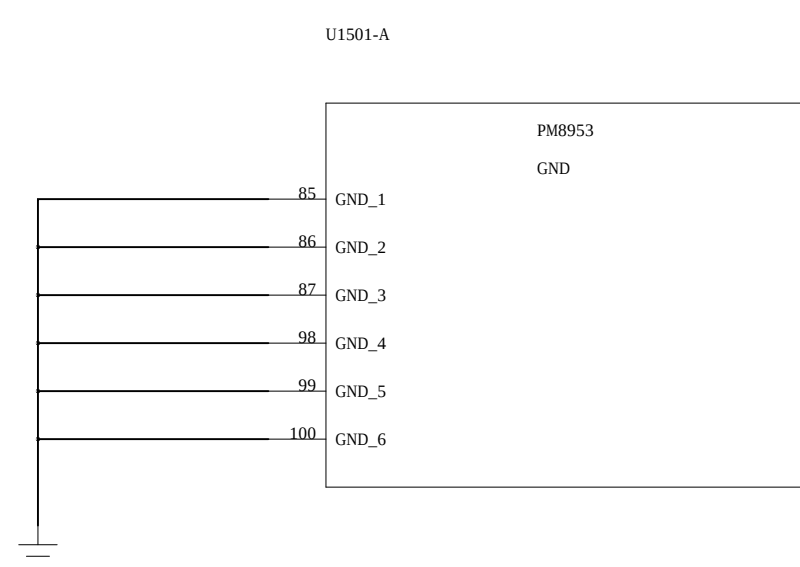
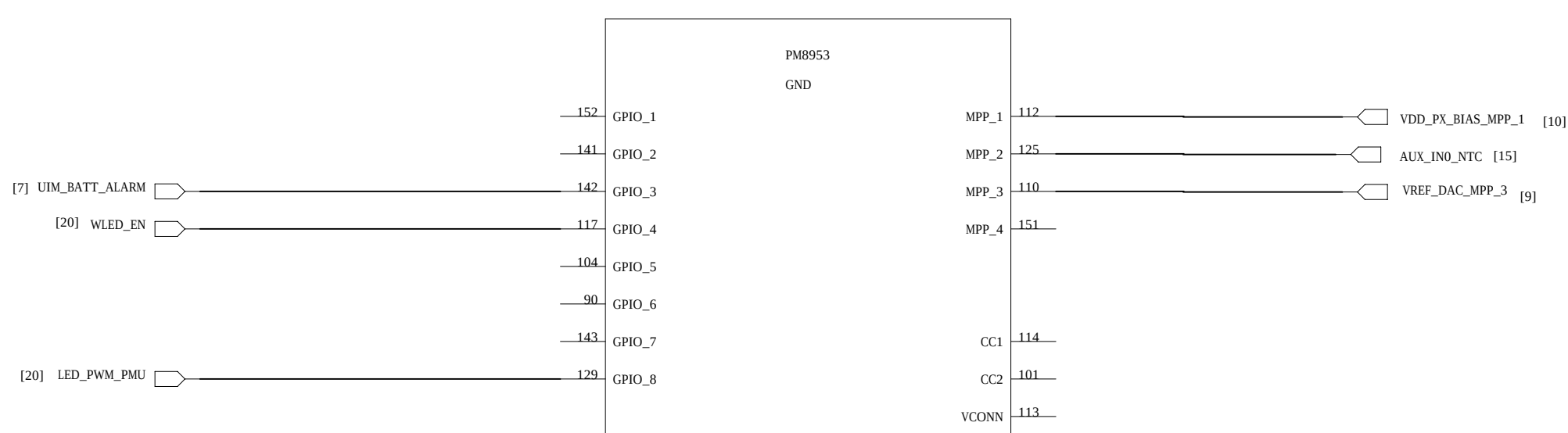
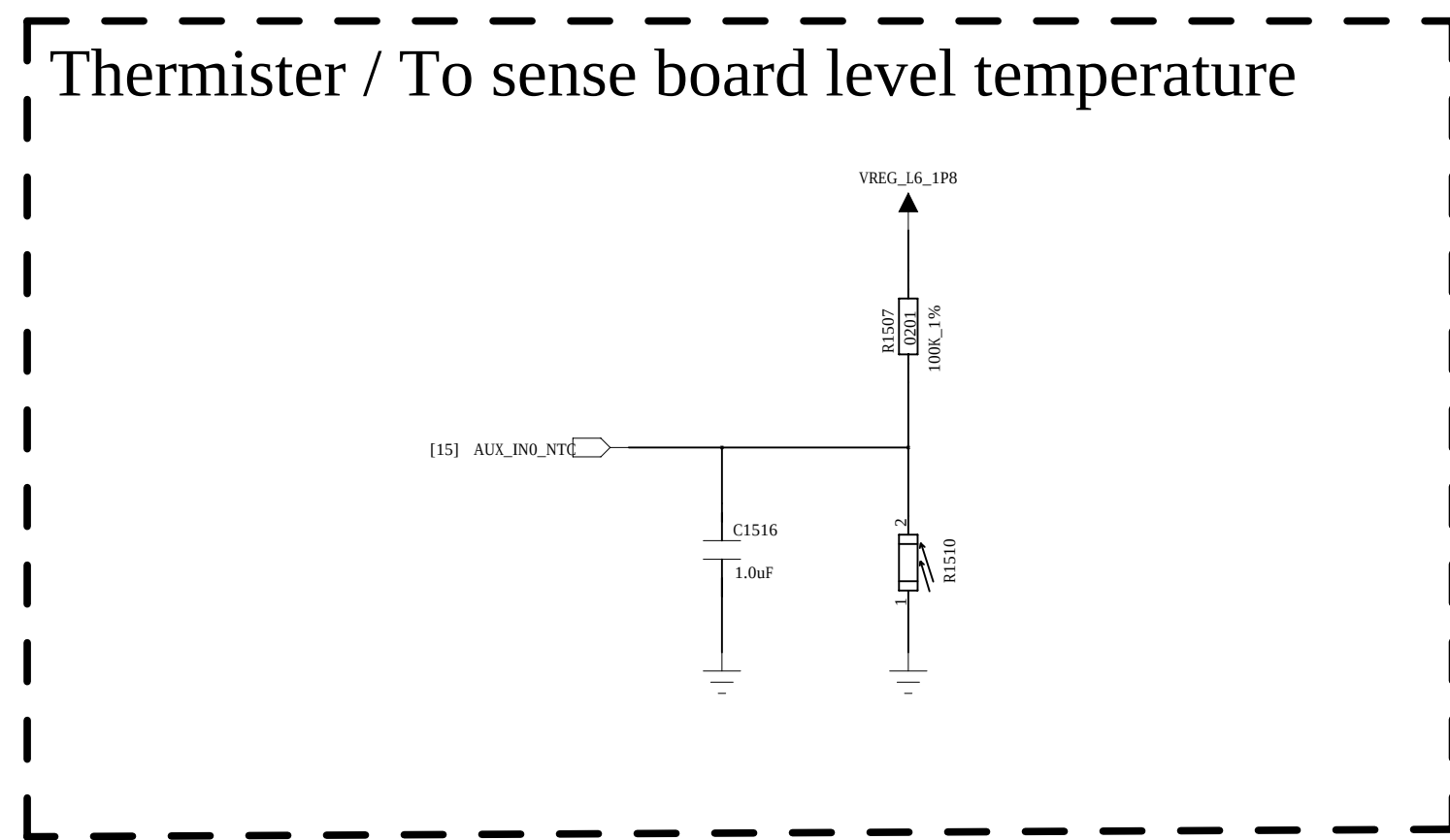
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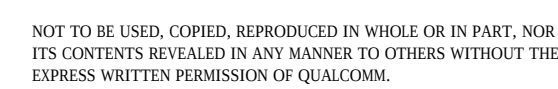
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MSM8953 GND





PM8953 Control/Interface



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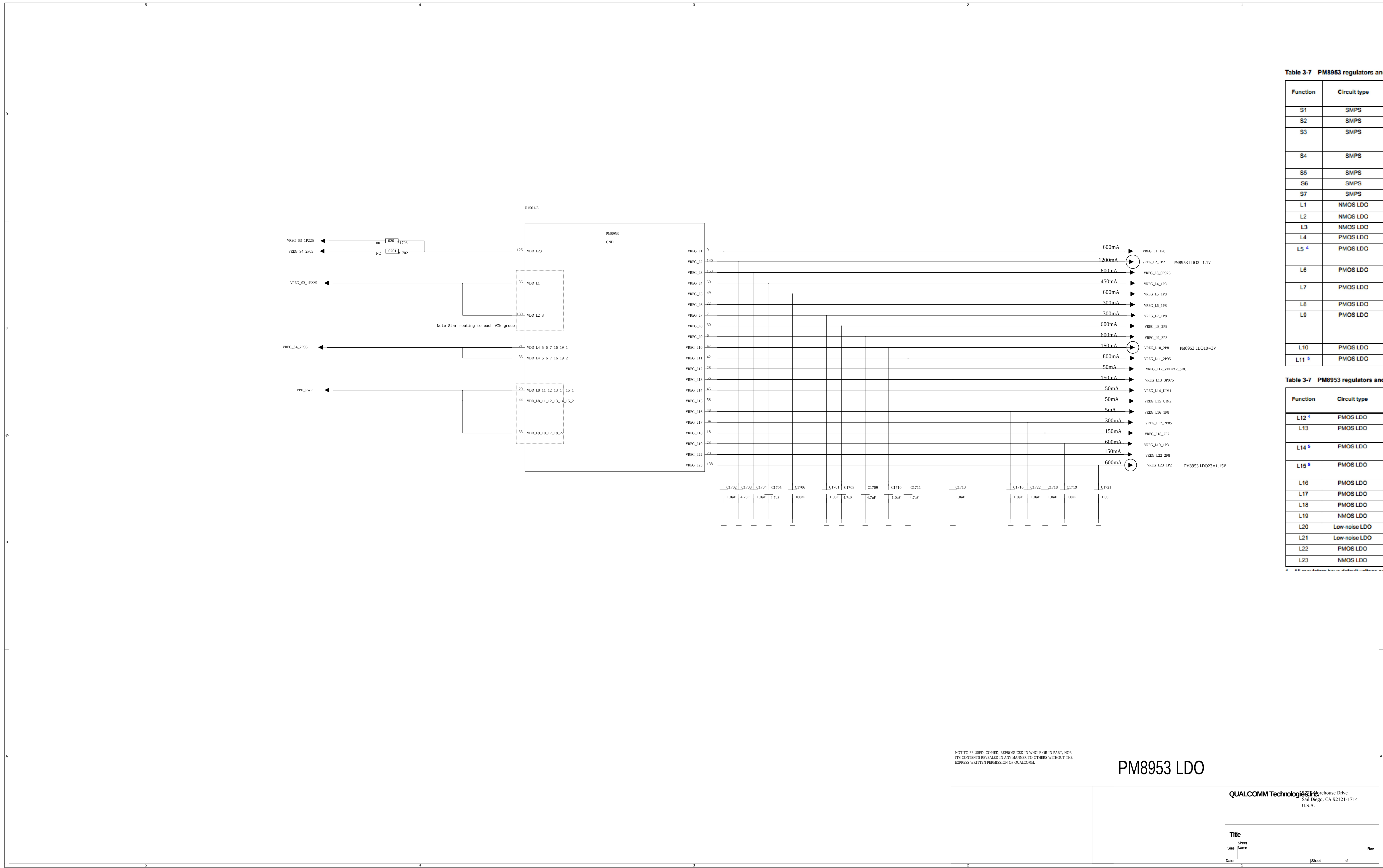


Table 3-7 PM8953 regulators and their intended uses

Function	Circuit type	Default voltage (V) ¹	Specified range (V) ¹ (MSM8953/SDM450)	Programmable range (V)	Rated current (mA)	Default on	Expected use (MSM8953/SDM450)	
S1	SMPS	0.87	0.4–1.14	0.32–2.04	3000	N	MSM modem	
S2	SMPS	0.87	0.4–1.14	0.32–2.04	4000	Y	MSM core and graphics	
S3	SMPS	1.225	1.2–1.25	0.32–2.04	2000	Y	LPDDR2 and LPDDR3, MIPI CSI, and CSI. Low-voltage LDOs (1, 2, 3, and 23)	
S4	SMPS	2.04	1.8–2.04	0.32–2.04	2000	Y	High-voltage LDOs (4, 5, 6, 7, 16, 19 RFCLK, and XO)	
S5	SMPS	0.87	0.4–1.14	0.350–1.355	3750	Y	MSM applications processor	
S6	SMPS	0.87	0.4–1.14	0.350–1.355	3750	Y	MSM applications processor	
S7	SMPS	0.915	0.4–1.14	0.32–2.04	2000	Y	MSM VDD memory rail (VDDMX)	
L1	NMOS LDO	1.0 ³	1.0 ³	0.375–1.5375	600	N	RFICs	
L2	NMOS LDO	1.100	1.100	0.375–1.5375	1200	N	Camera: digital	
L3	NMOS LDO	0.925	0.925	0.375–1.5375	600	Y	MSM DSI PLL and USB	
L4	PMOS LDO	1.800	1.800	1.750–3.3375	450	N	RFICs and GPS eLNA	
L5 ⁴	PMOS LDO	1.800	1.800	1.750–3.3375	600	Y	Most digital ICs, MSM pad groups 3 and 7, LPDDR, and eMMC	
L6	PMOS LDO	1.800	1.800	1.750–3.3375	300	N	MSM QFPROM, camera, touchscreen, display, and sensors	
L7	PMOS LDO	1.800	1.800	1.750–3.3375	300	Y	MSM analog, USB and PLLs, WCN XO, and PM baseband clock driver	
L8	PMOS LDO	2.900	2.900	1.750–3.3375	600	Y	eMMC	
L9	PMOS LDO	$V_{out} = 3.3\text{ V for } V_{BAT} > 3.575\text{ V}$ $V_{out} = 3\text{ V for } V_{BAT} < 3.575\text{ V}$		3.000–3.300	1.750–3.3375	600	N	WCN
L10	PMOS LDO	3.0	3.0	1.750–3.3375	150	N	Sensors and touchscreen	
L11 ⁵	PMOS LDO	2.950	2.950	1.750–3.3375	800	Y	Micro SD	

Table 3-7 PM8953 regulators and their intended uses (cont.)

Function	Circuit type	Default voltage (V) ¹	Specified range (V) ² (MSM8953/SDM450)	Programmable range (V)	Rated current (mA)	Default on	Expected use (MSM8953/SDM450)
L12 ⁴	PMOS LDO	2.950	1.800/2.950	1.750–3.3375	50	Y	MSM pad group 2
L13	PMOS LDO	3.125	3.125	1.750–3.3375	150	Y	MSM USB and PMIC and external codec audio
L14 ⁵	PMOS LDO	1.800	1.800/3	1.750–3.3375	50	N	MSM pad group 5, dual-voltage UIM1, and NFC
L15 ⁵	PMOS LDO	1.800	1.800/3	1.750–3.3375	50	N	MSM pad group 6 and dual-voltage UIM2
L16	PMOS LDO	1.800	1.800	1.750–3.3375	5	N	PMIC HKADC
L17	PMOS LDO	2.850	2.850	1.750–3.3375	300	N	Camera and display
L18	PMOS LDO	2.700	2.700	1.750–3.3375	150	N	QTI RF front-end
L19	NMOS LDO	1.350	1.350	0.375–1.5375	600	N	MSM analog, WCN, and WGR
L20	Low-noise LDO	1.74	1.74	1.74–3.3375	5	Y	PMIC XO circuits
L21	Low-noise LDO	1.74	1.74	1.74–3.3375	5	Y	PMIC RF clock buffers
L22	PMOS LDO	2.800	2.800	1.750–3.3375	150	N	Camera: analog
L23	NMOS LDO	1.15	1.15	0.375–1.5375	600	N	Camera: digital

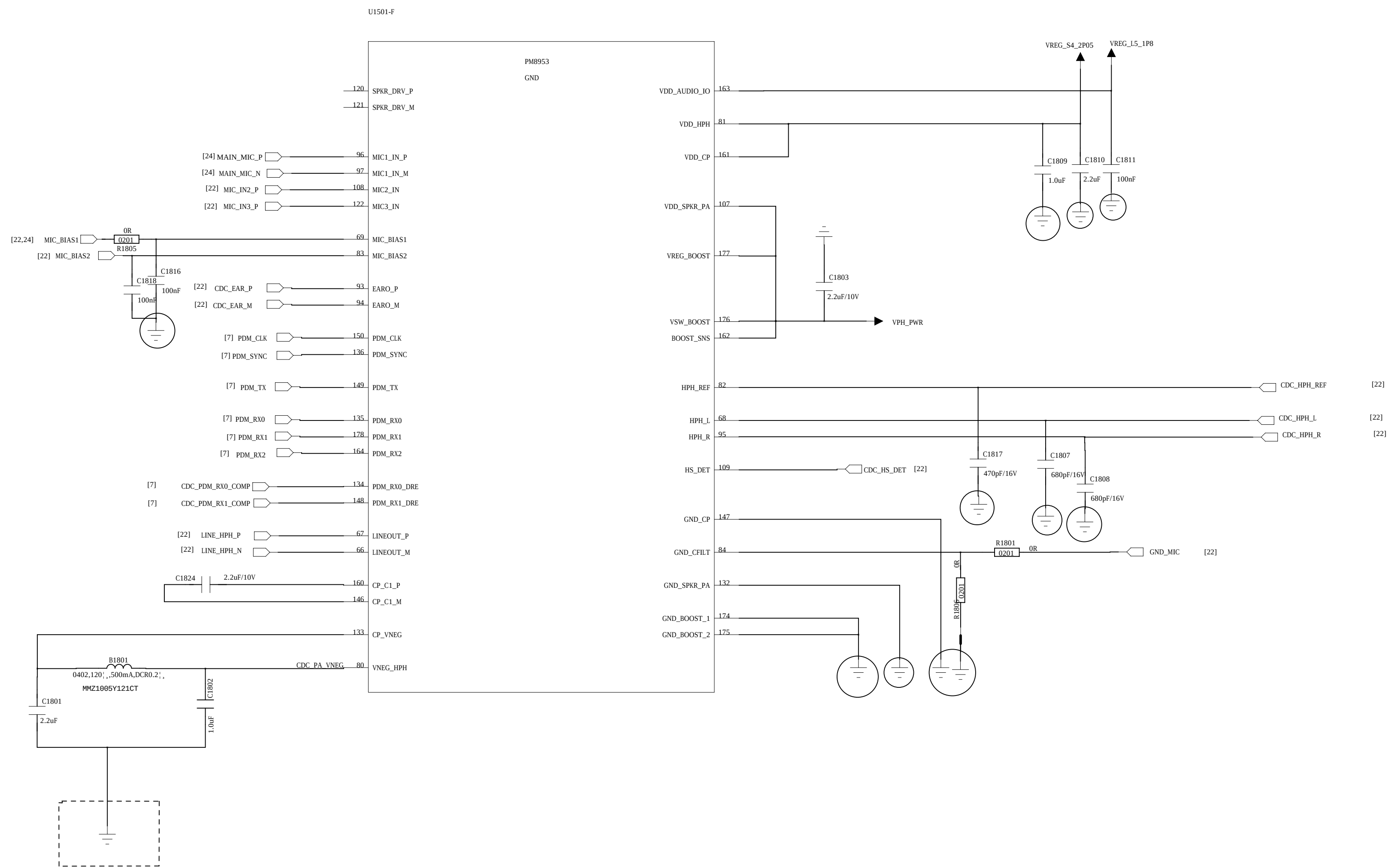
¹ All conclusions have default values, whether or not they default to the values and state denote from the recommended load capacitor (RLC) specification.

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PM8953 LDO

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Sheet
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Date



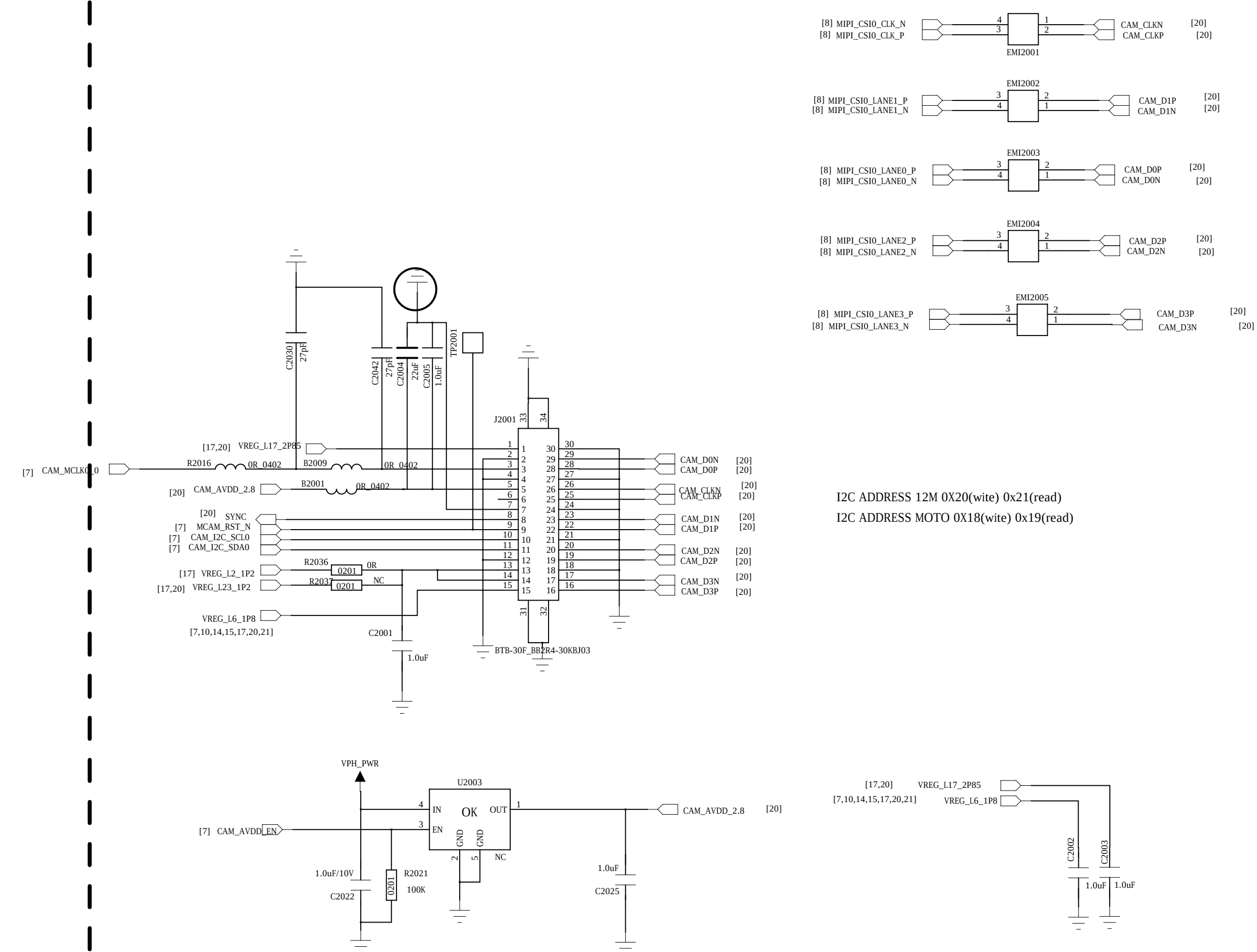
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PM8953 CODEC/Audio PA

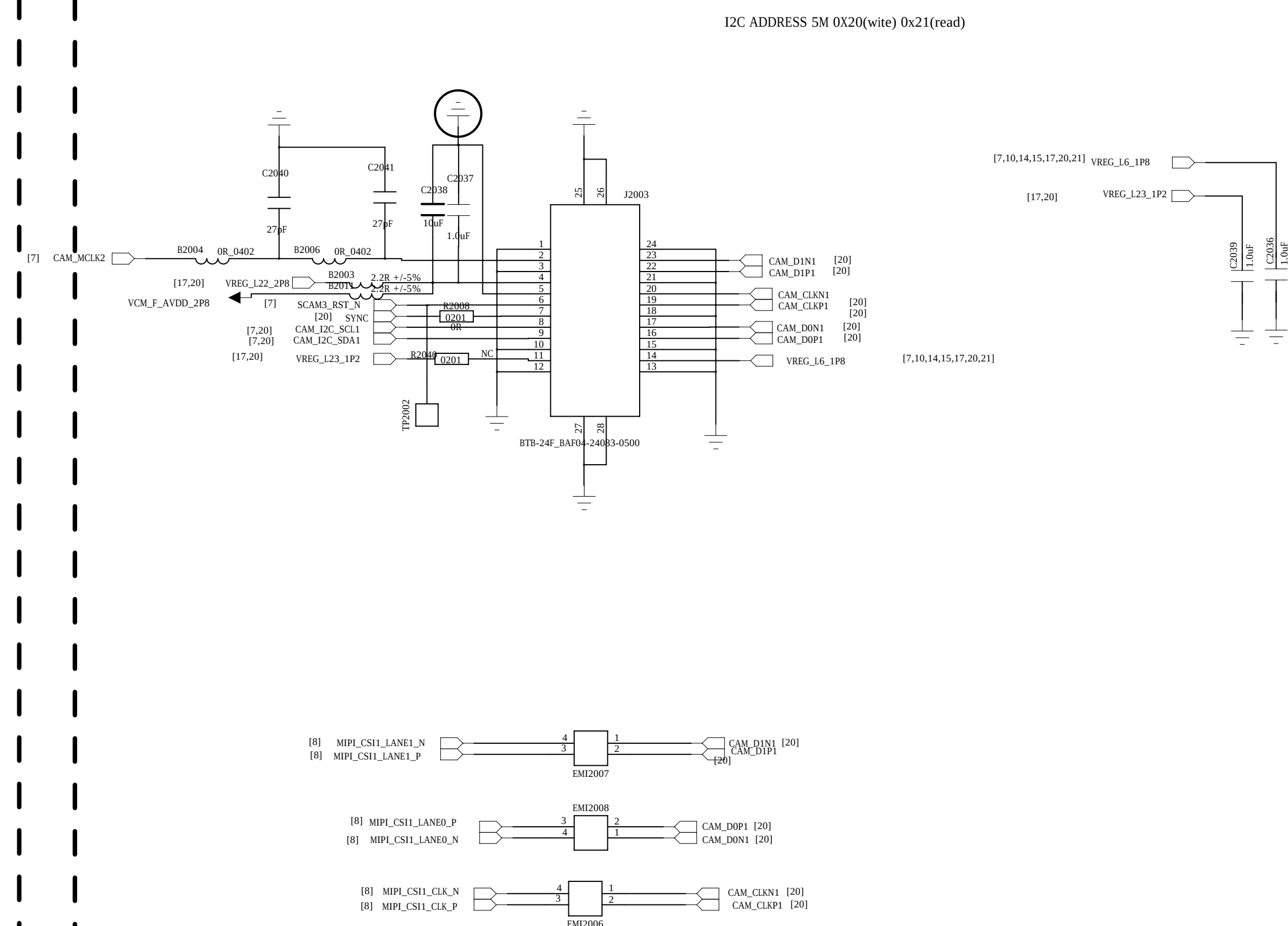
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REVISION RECORD			
LTB	ECO NO.	APPROVED	DATE

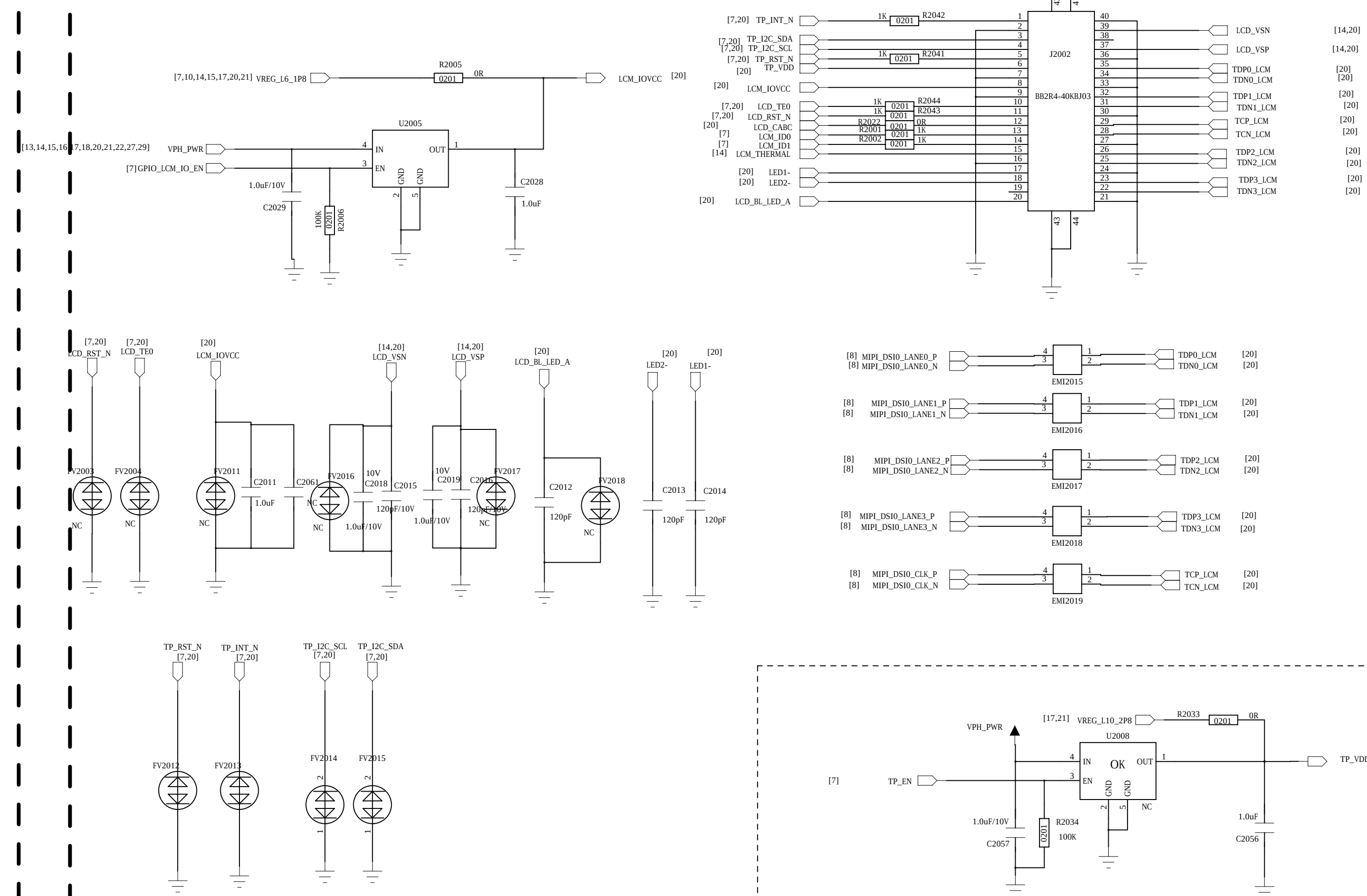
Main Camera A



Main Camera B

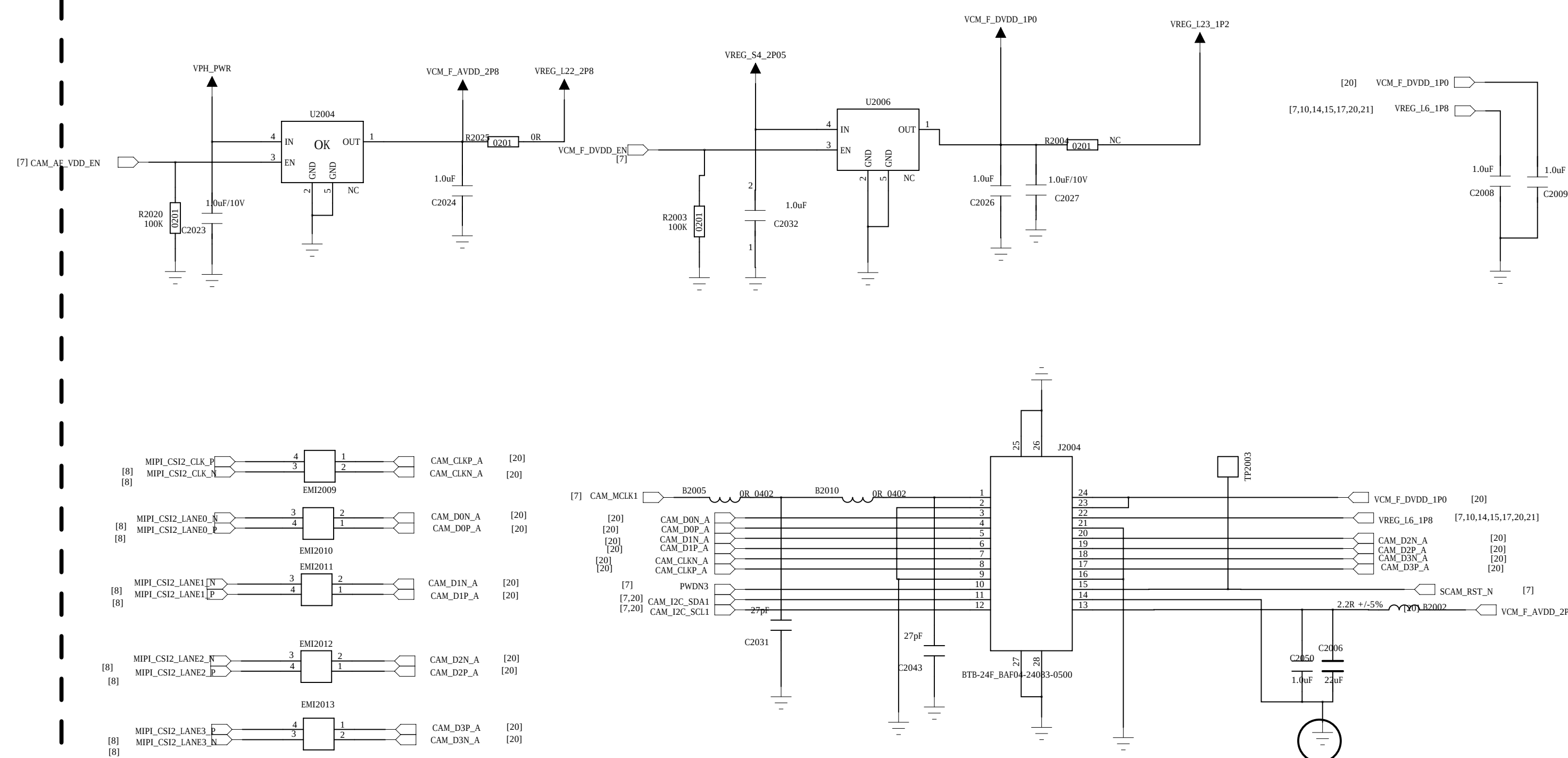


LCM

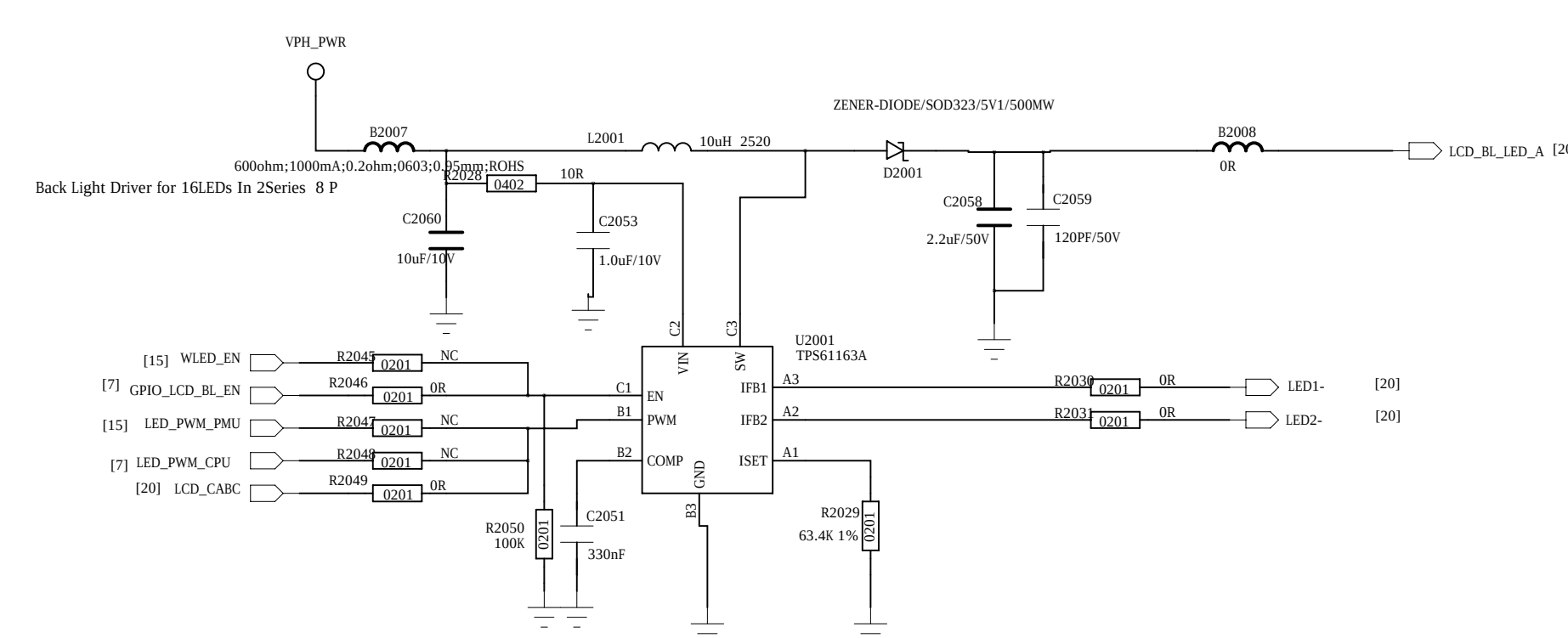


LDO

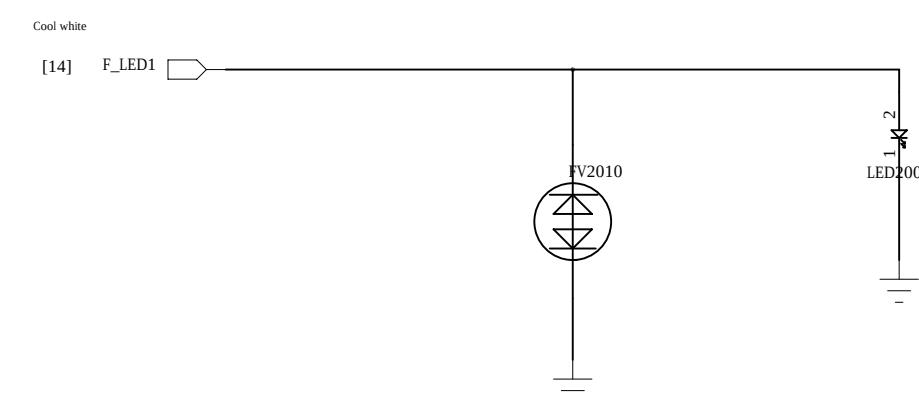
Front Camera



LCM Black Light



Main FLASHLIGHT



Schematic design notice of "63_PERI_CAMERA_KEYPAD" page.

Note 62-1: The VCC of I2C_0 is pulled to "VCAM_IO_PMU".

Note 62-2: I2C control interface of front camera (with AF) must be assigned to I2C-2 bus when PIP/IV feature be supported.

Note 62-3: Reserve a capacitor (27pF) on camera's MCLK and shunt it to GND to prevent GPS de-sense.

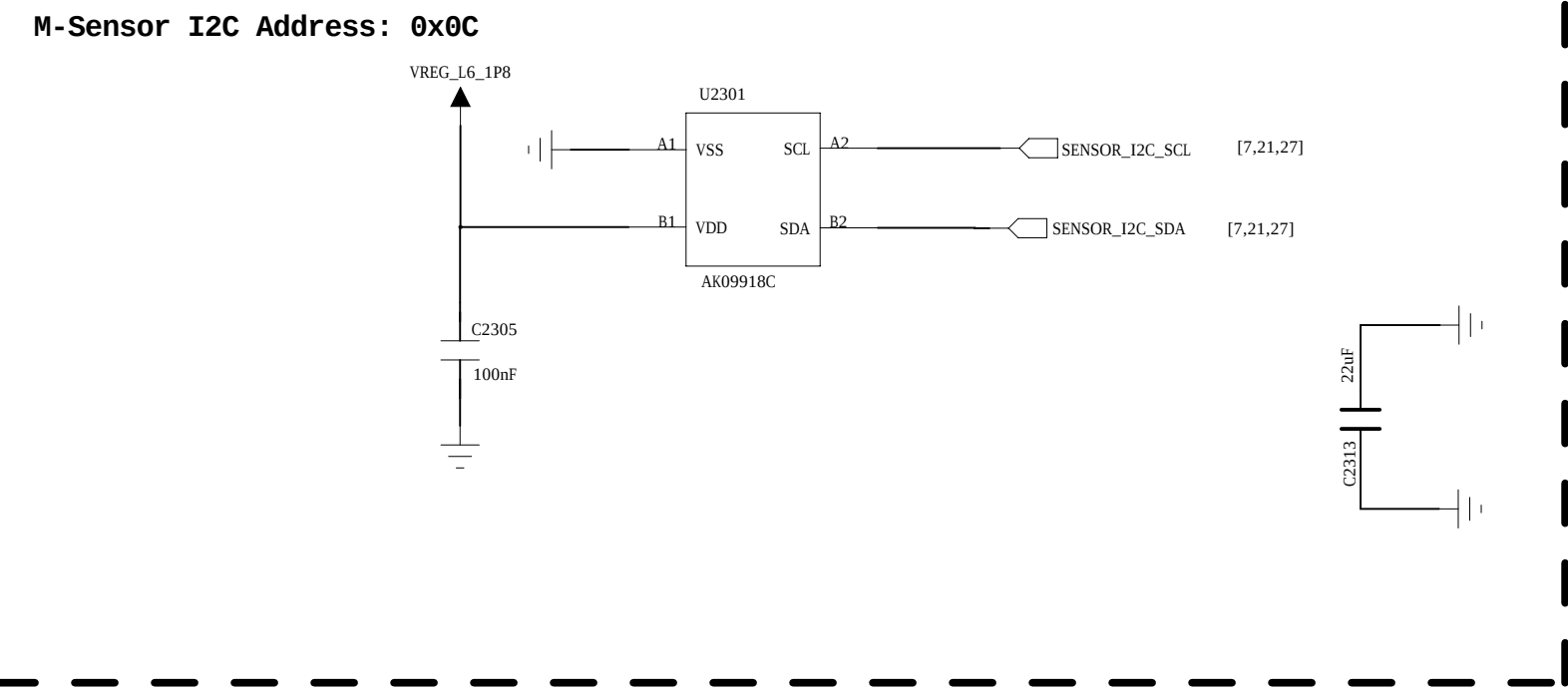
IC ADDRESS 2M 0x7a(wire) 0x7b(read)

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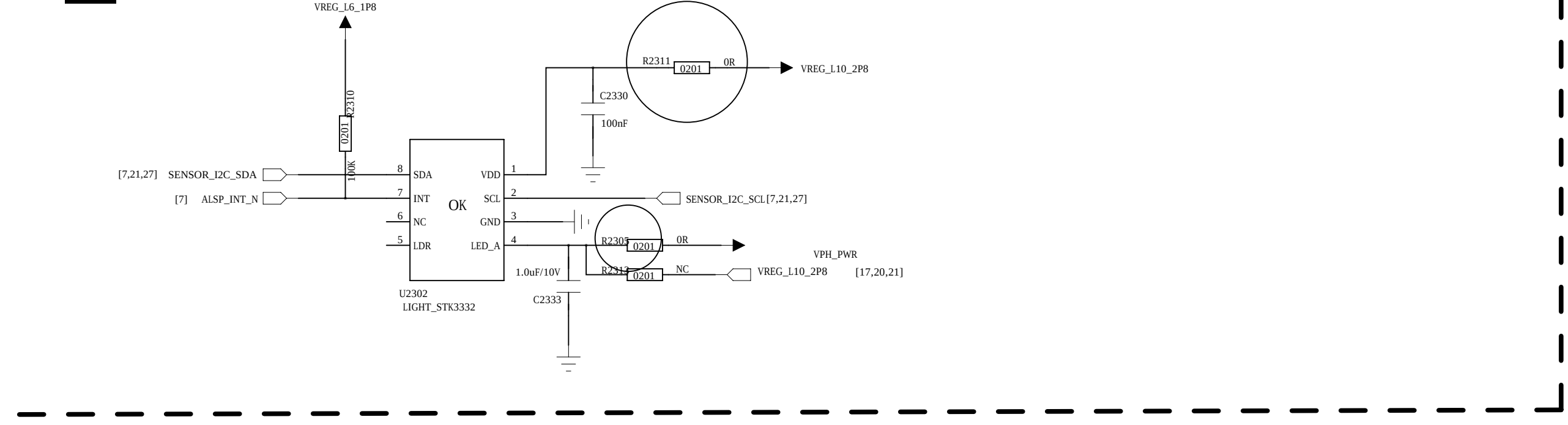
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L-SENSOR

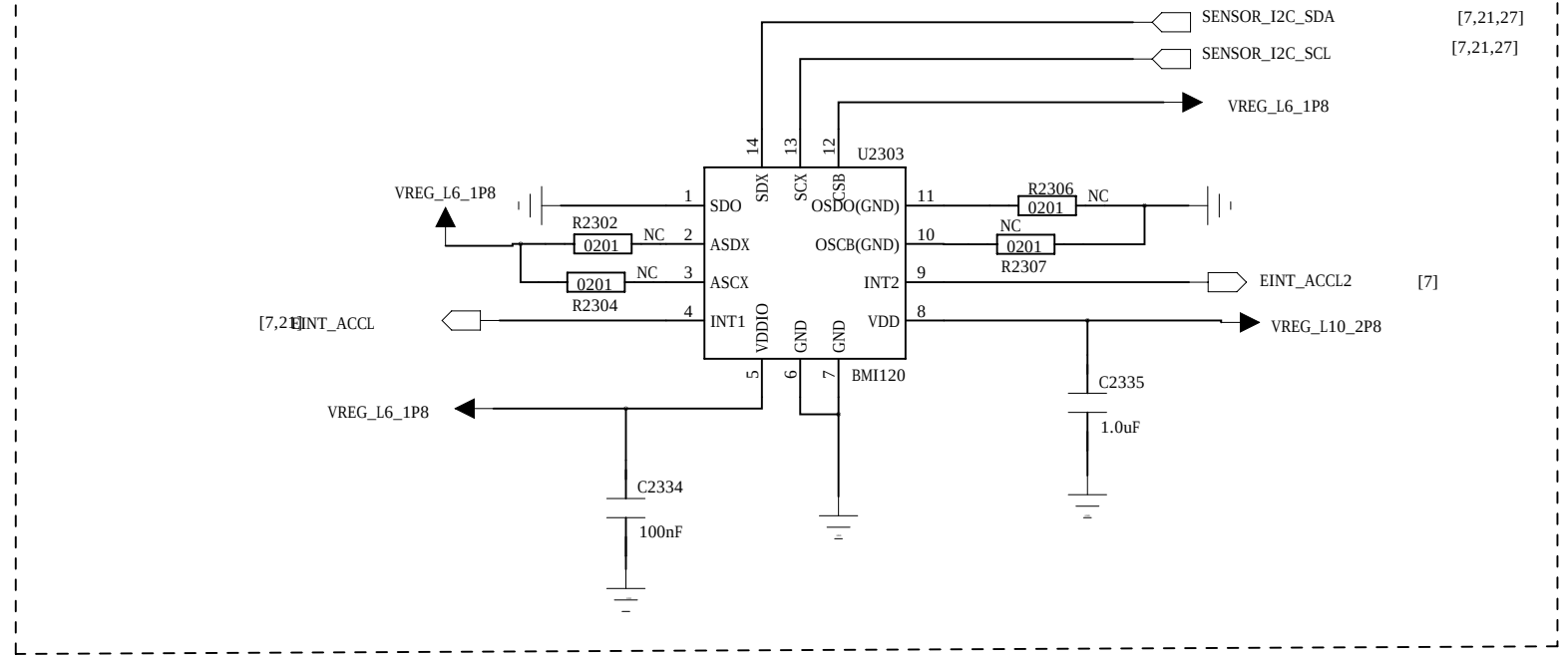
M-Sensor



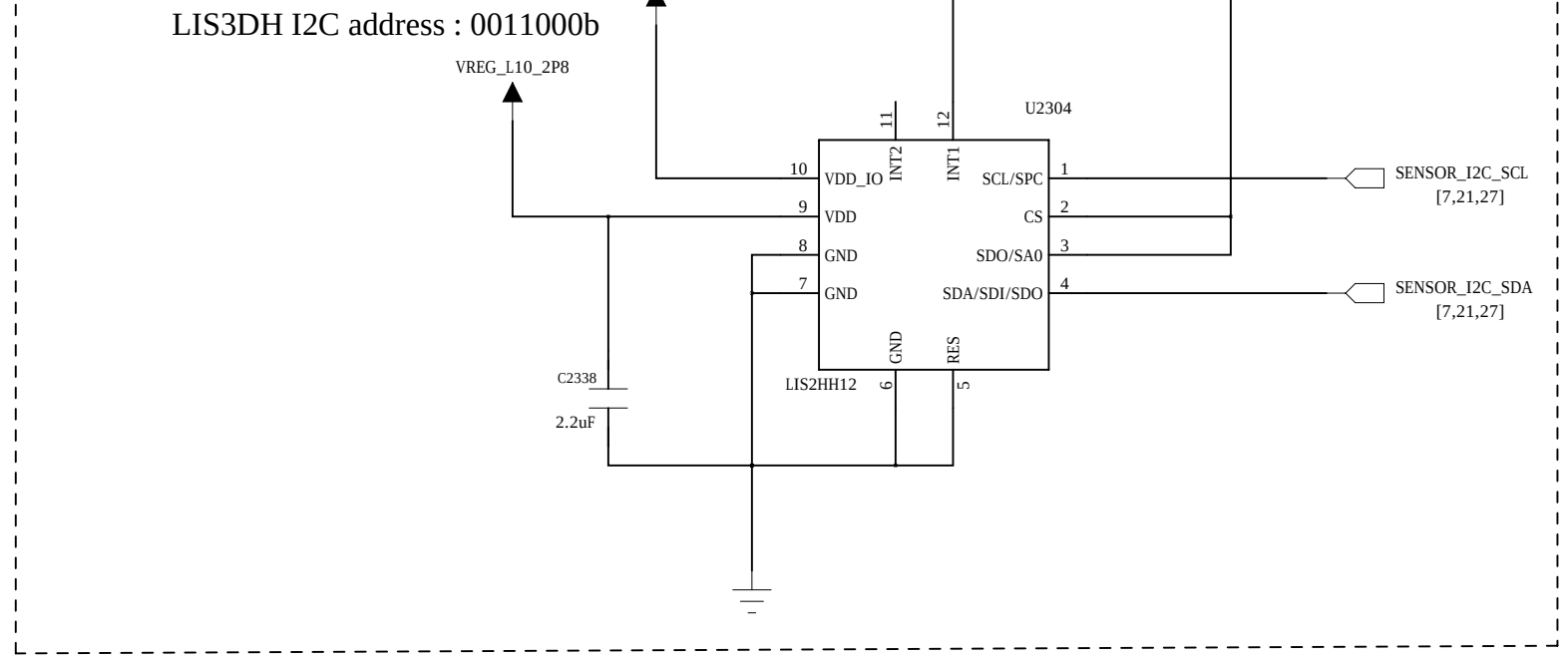
P_SENSOR



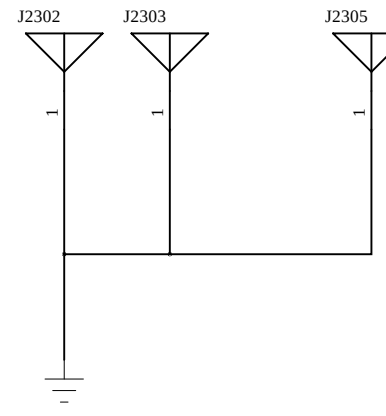
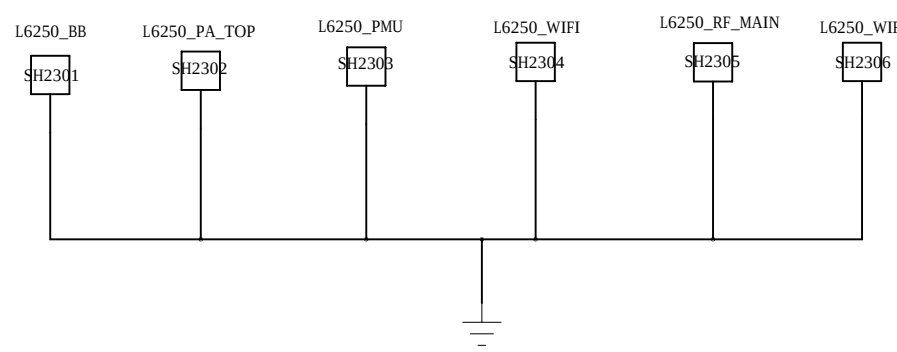
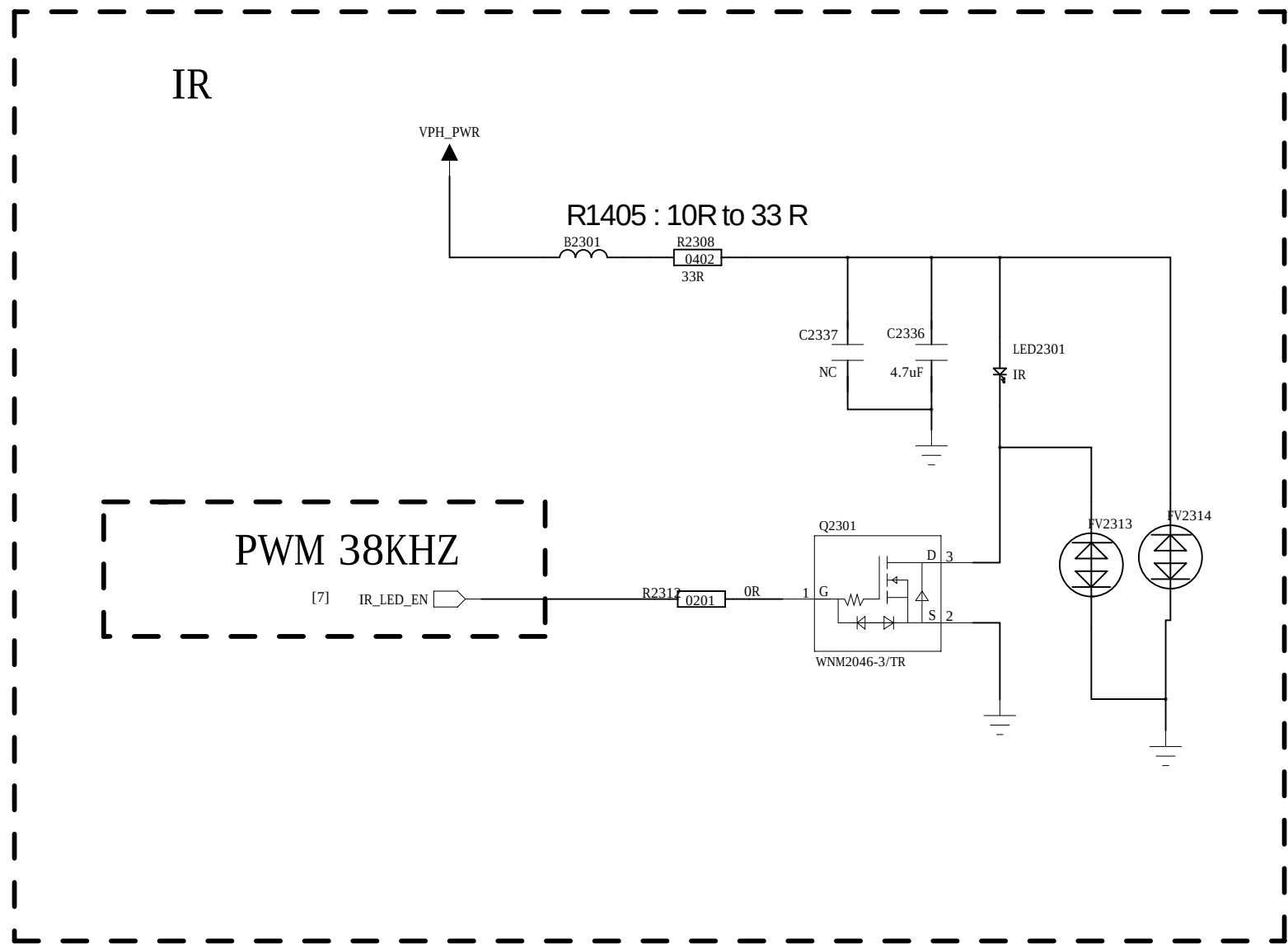
ACC-Sensor



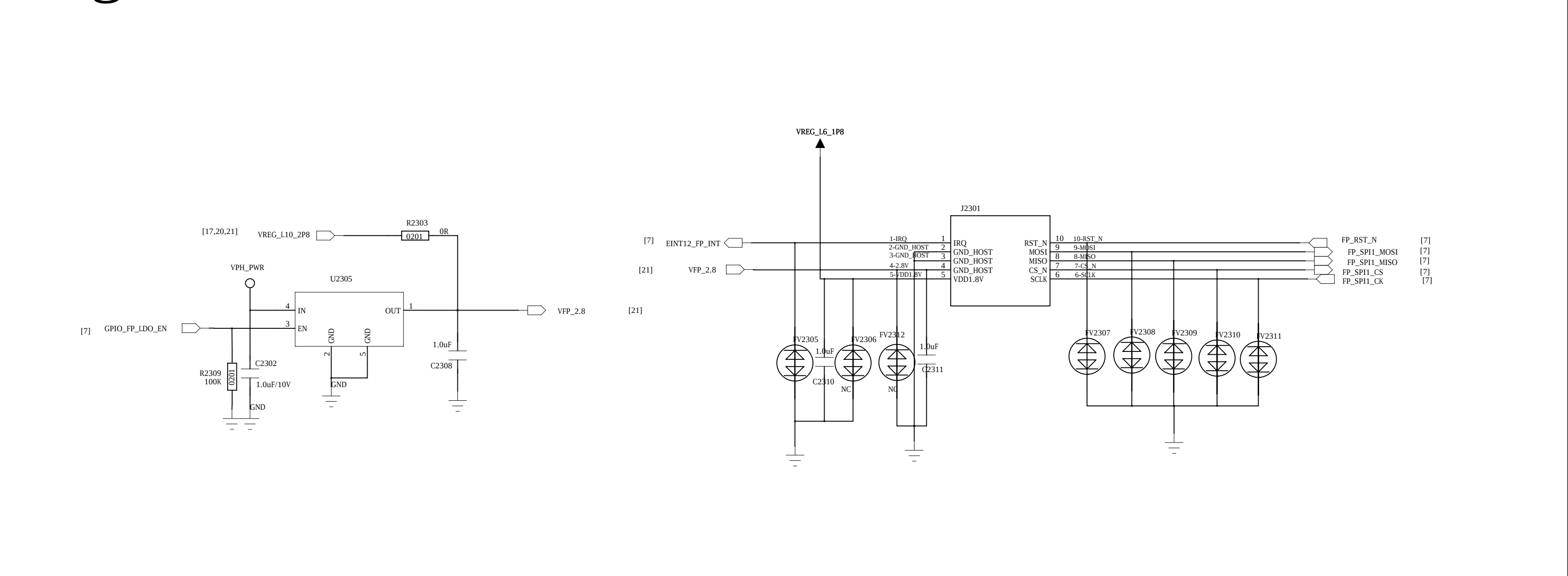
ACC-Sensor



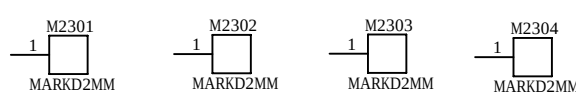
IR



FingerPrint-Sensor



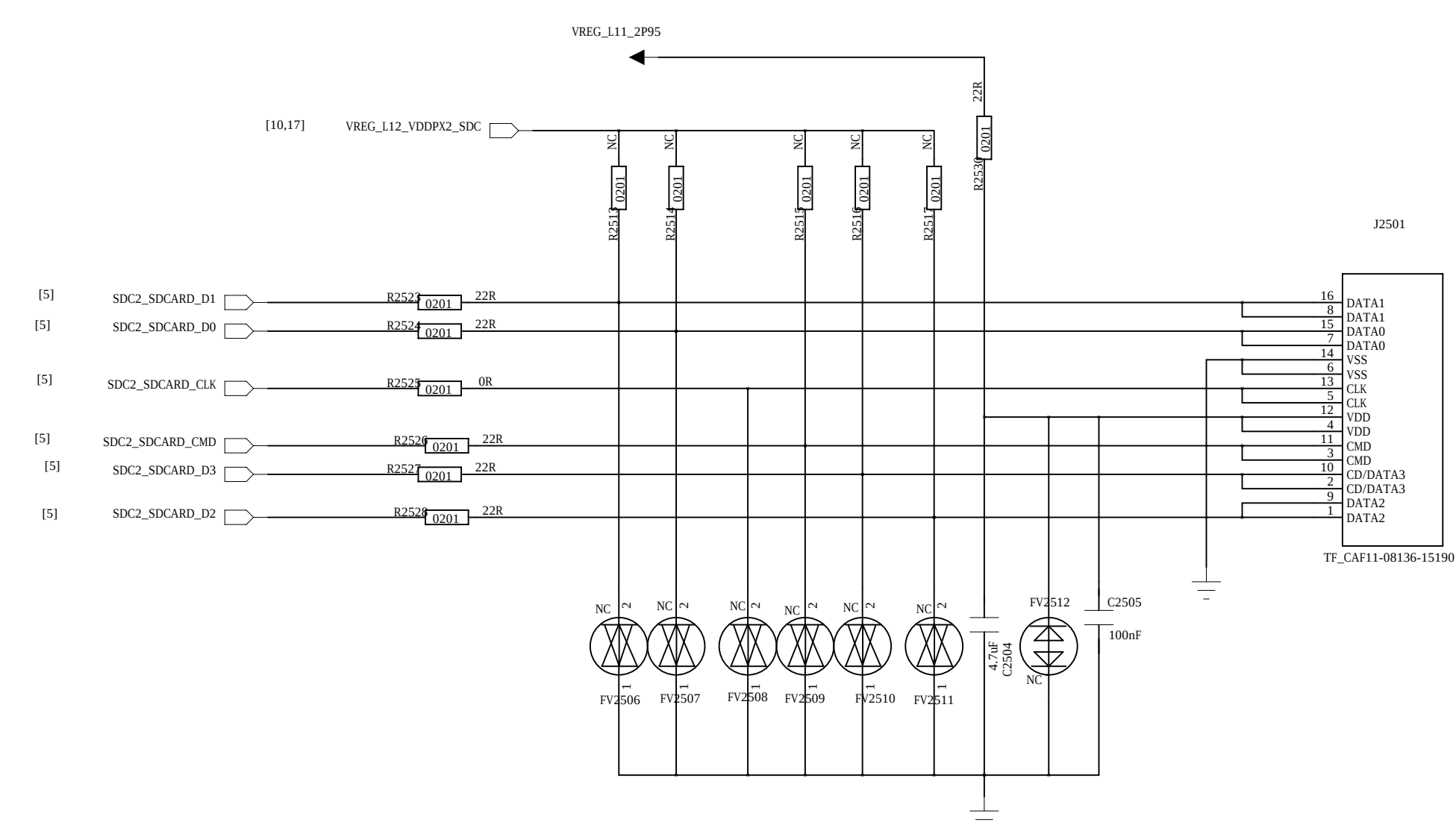
MARK POINT



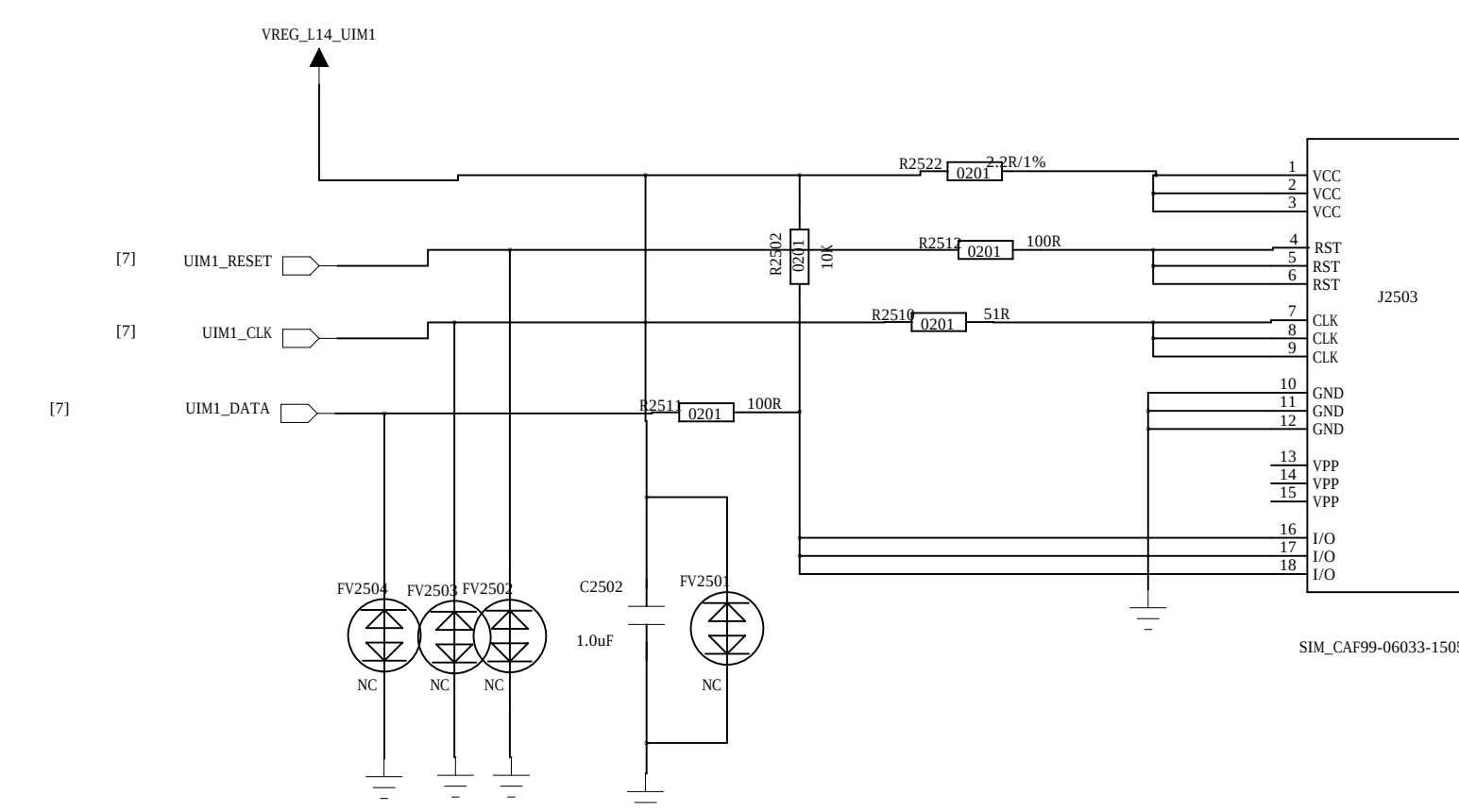
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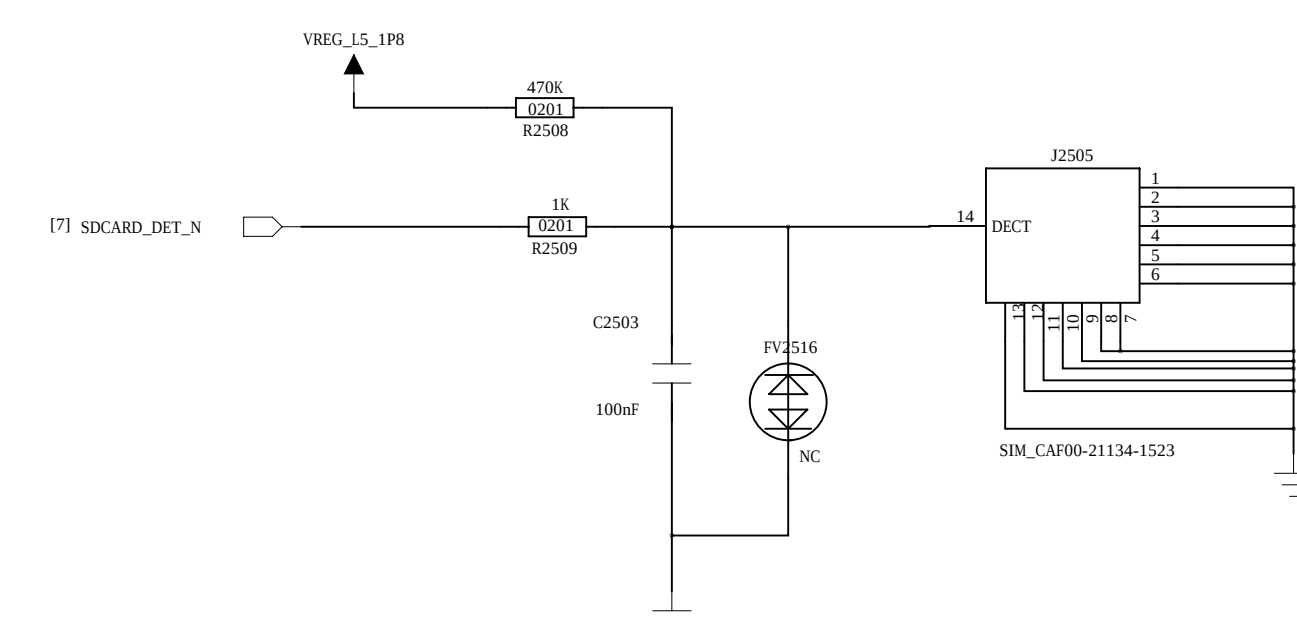
T-Card



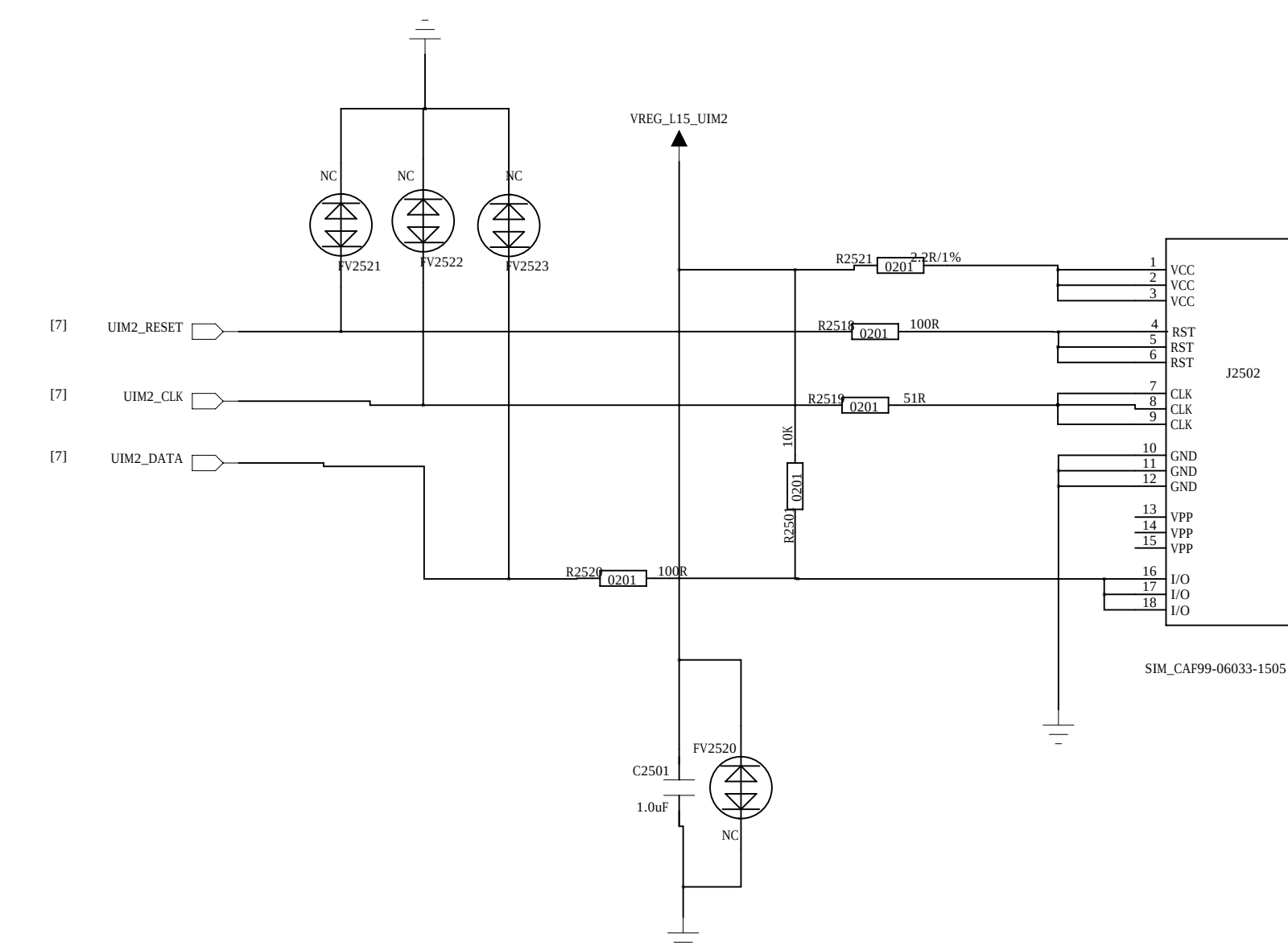
SIM1



INT

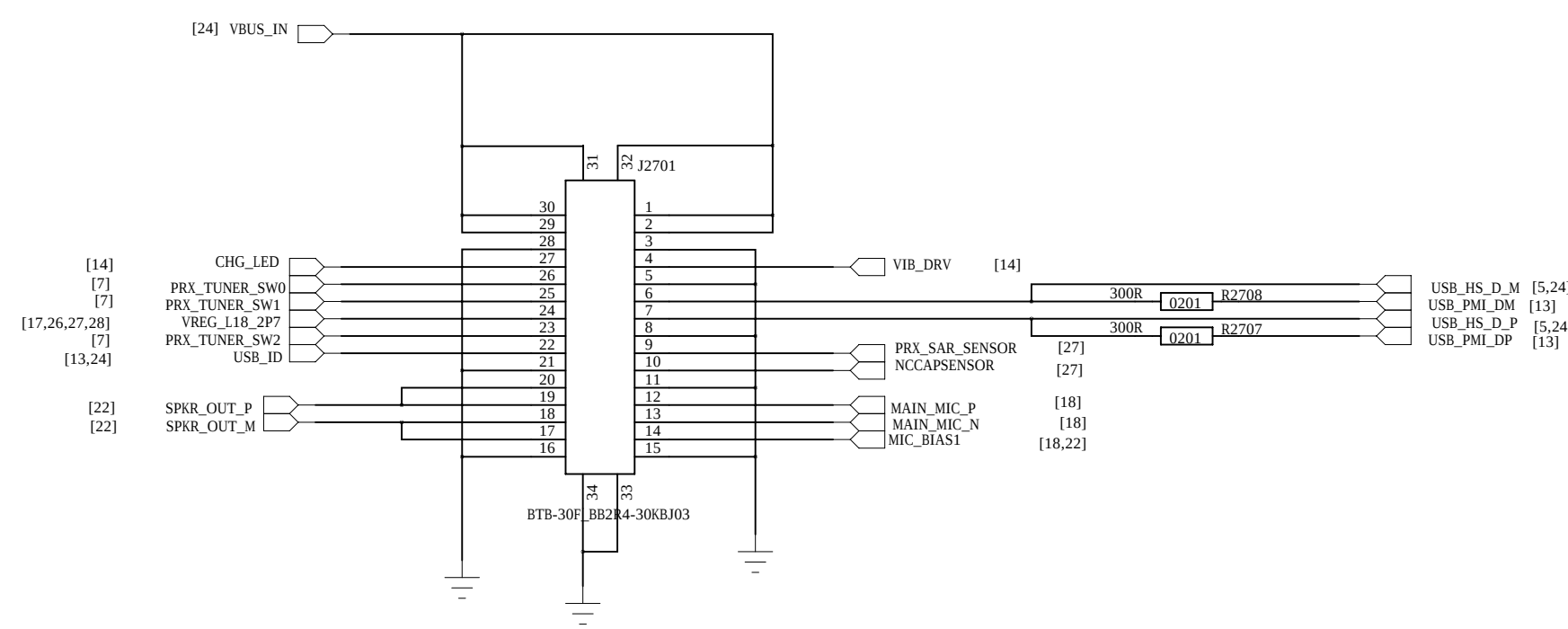


SIM2

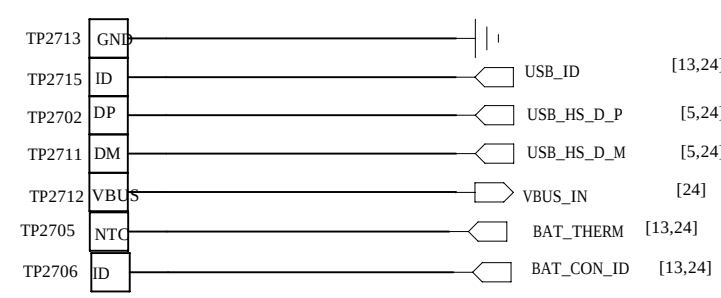


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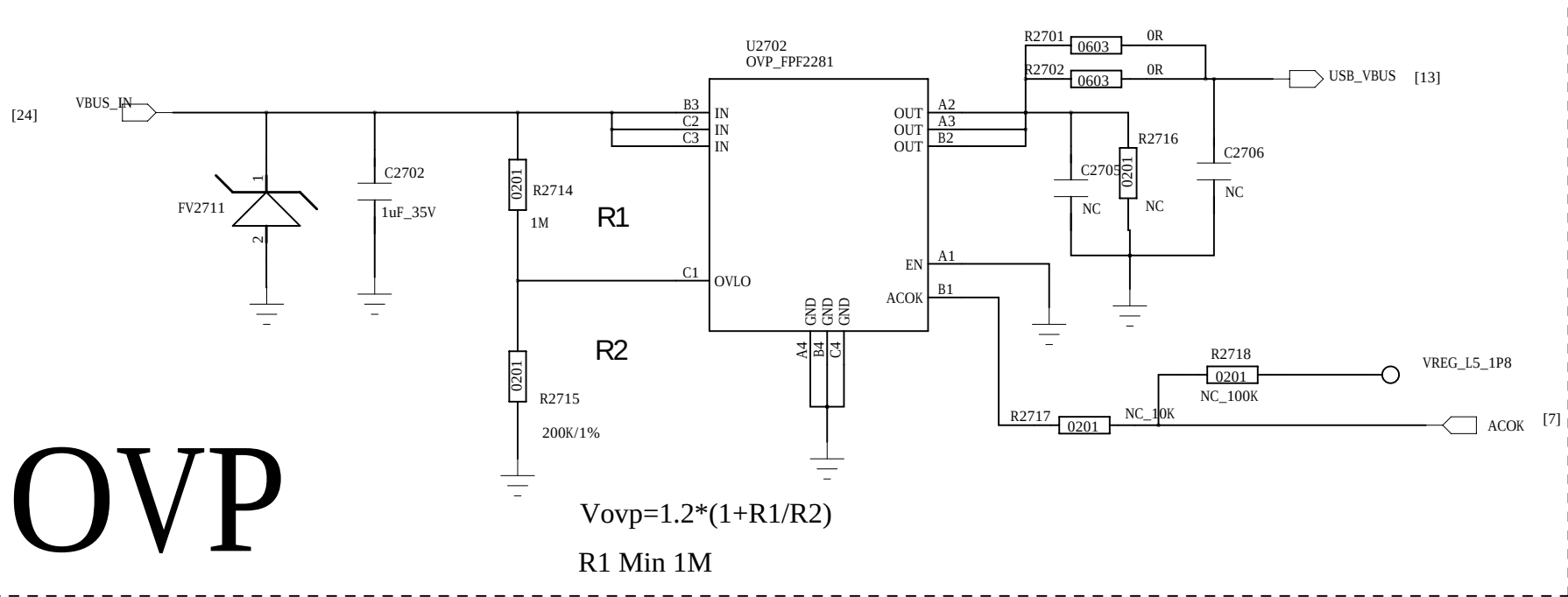
SUB_FPC



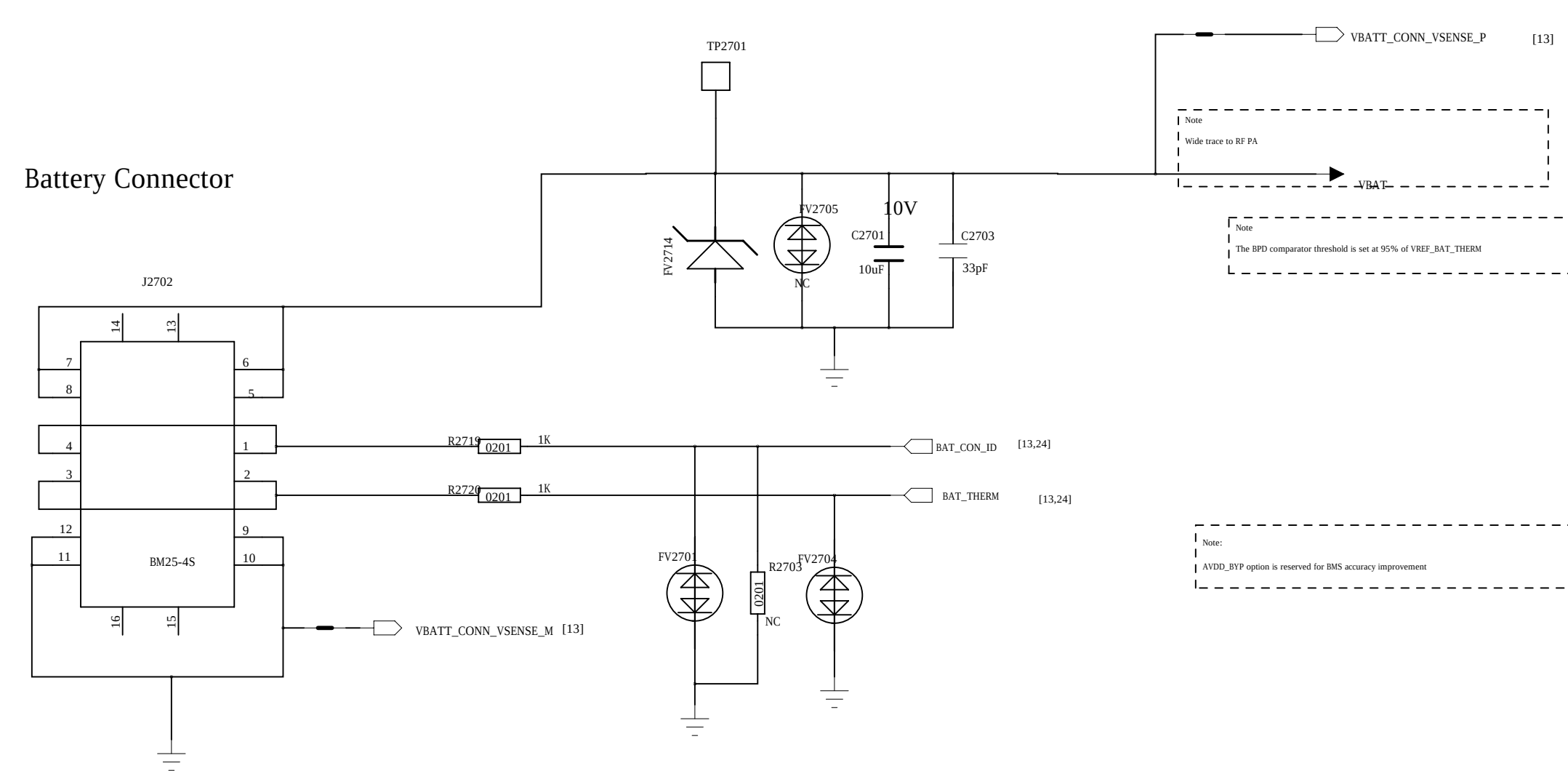
Test point



OVP

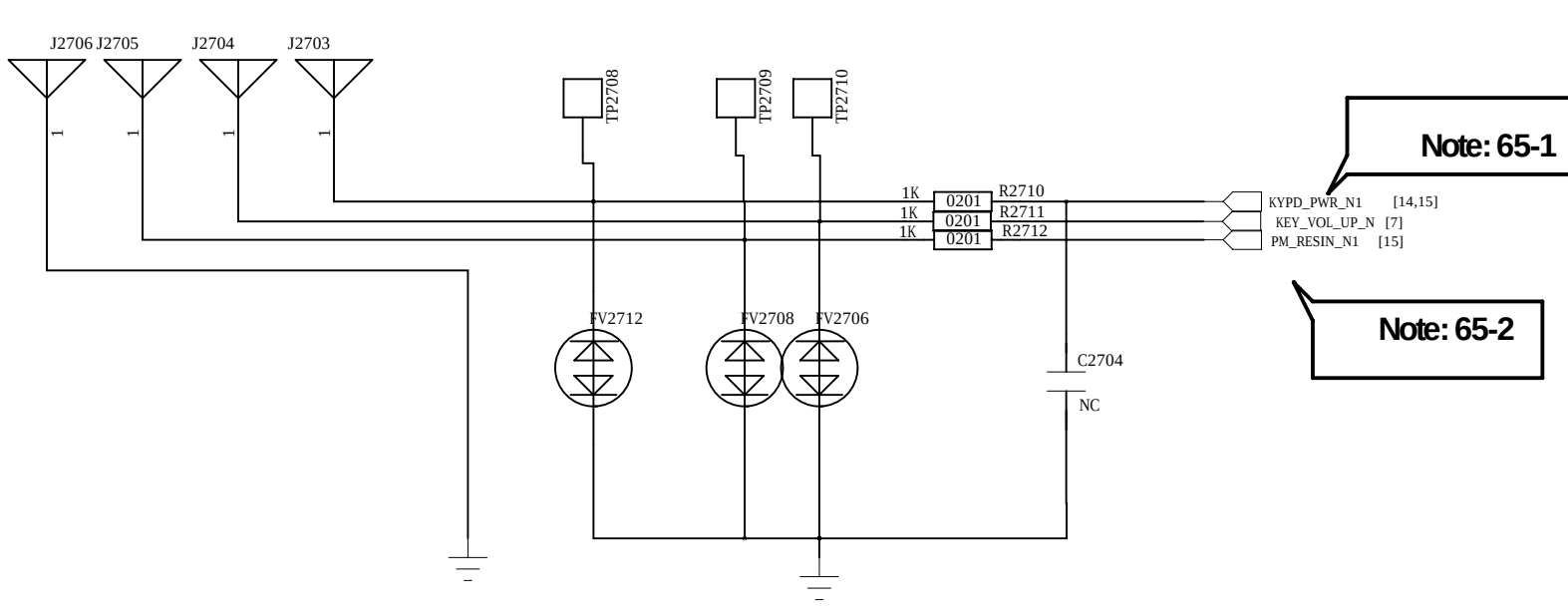


BAT



Note:
Battery ID resistor value require 20K~150K.
Add 100K to ground if battery package ID not meet requirement

Power Key

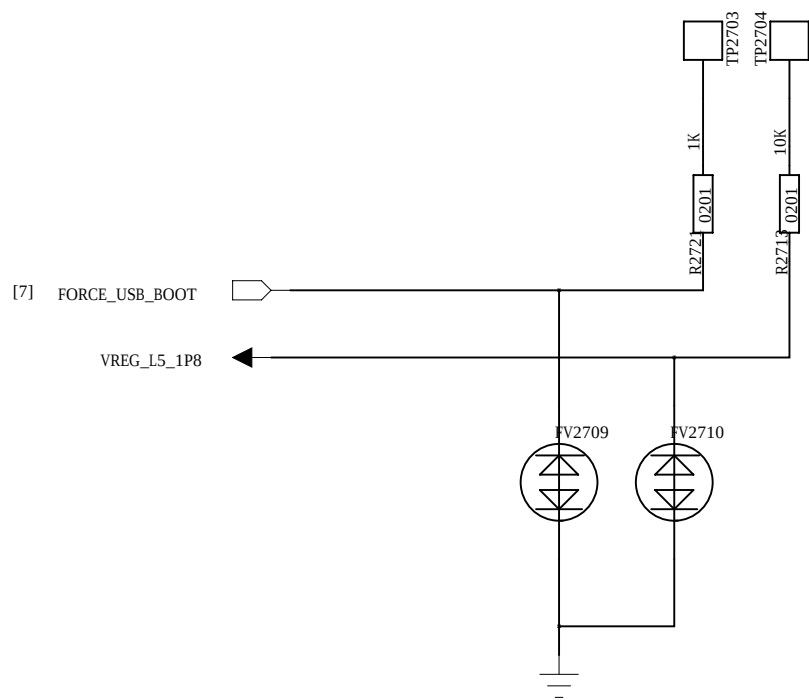


Schematic design notice of "65_PERI_Dual_SIM_ICUSB_KEYPAD" page.

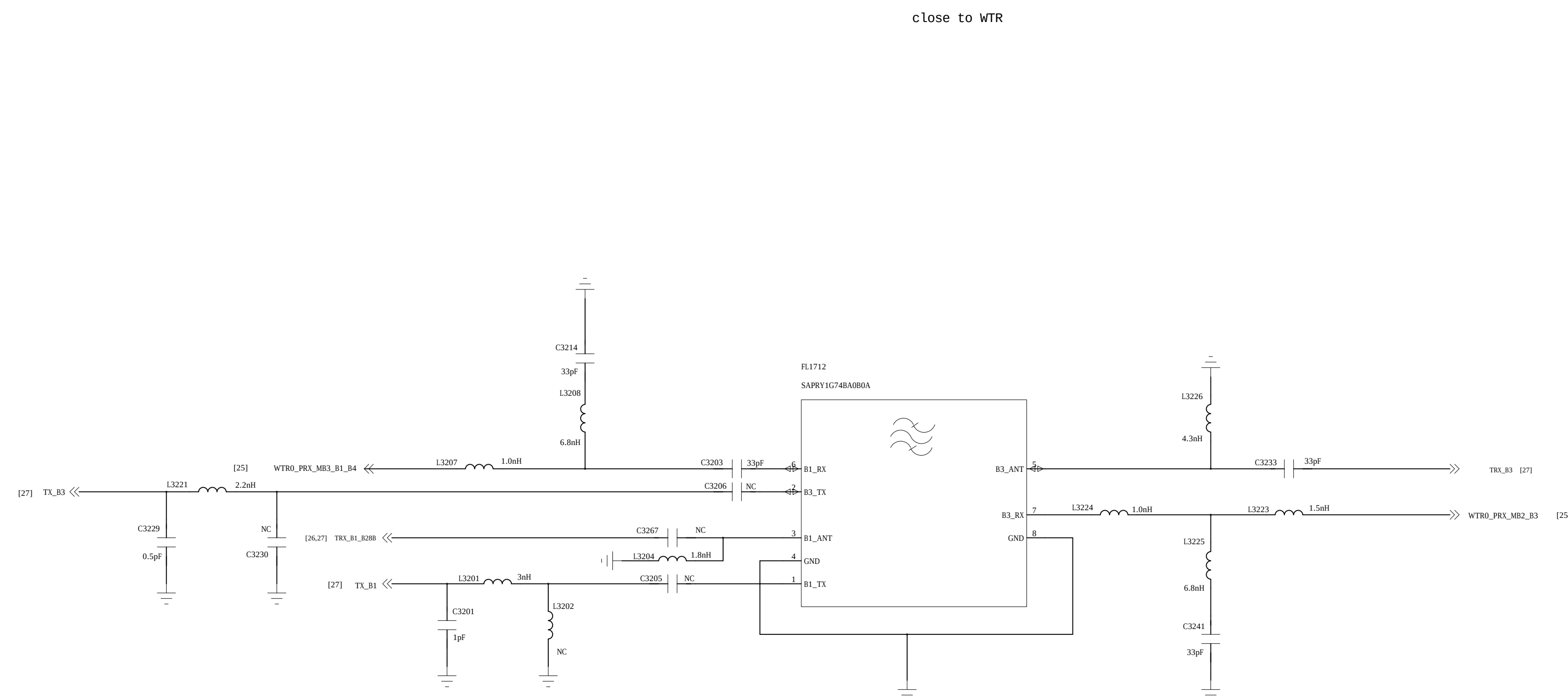
Note 65-1: DO NOT put pull-up resistor on PWRKEY

Note 65-2: Volume Up : HOME Key / GND
Volume Down : (KPROW0/KPCOL0) or KPCOL0 / GND

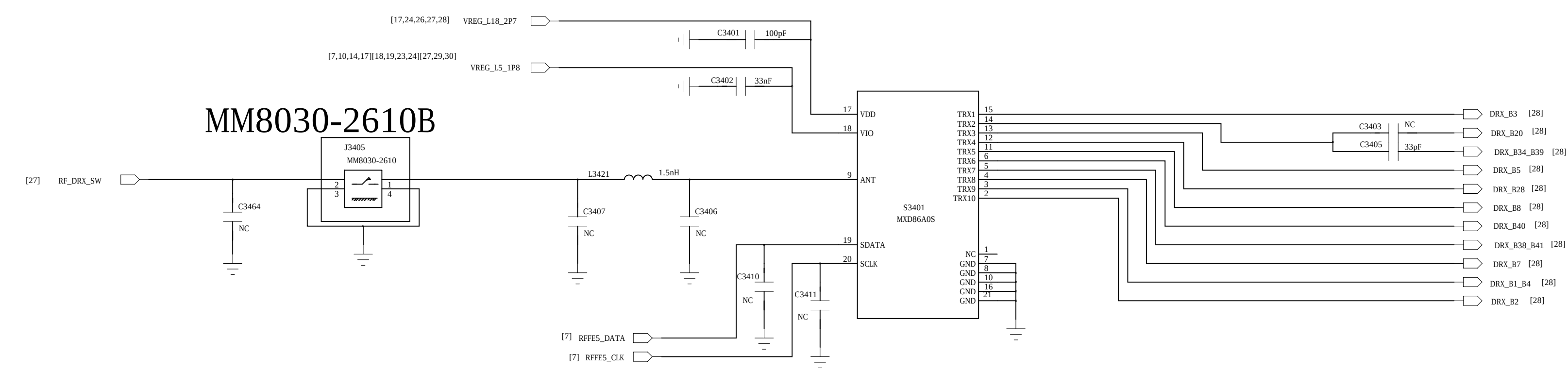
Force USB boot



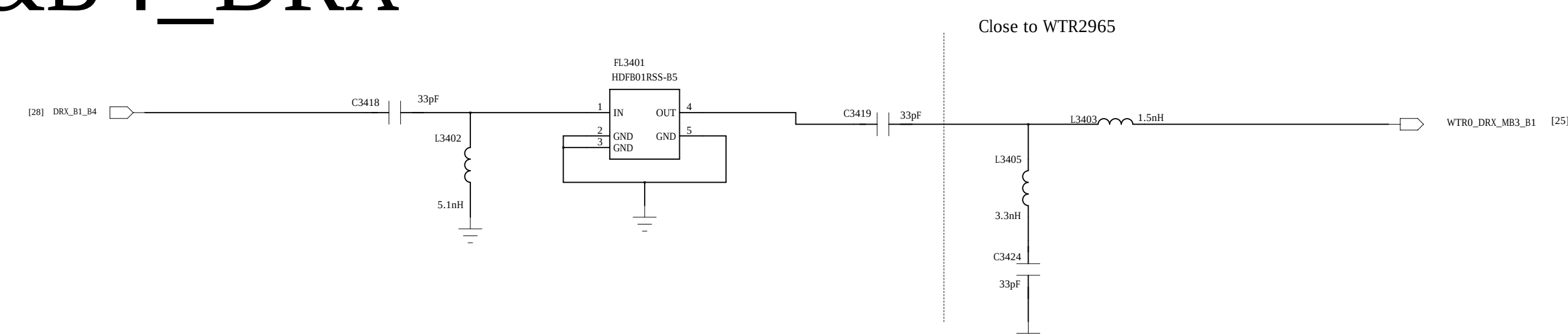
TRX_B1 TRX_B3



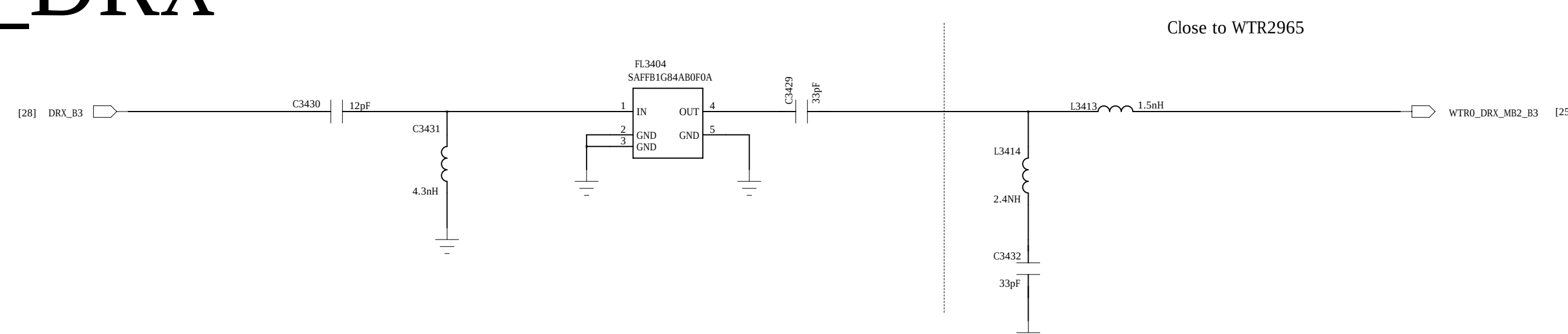
- ANT-Switch



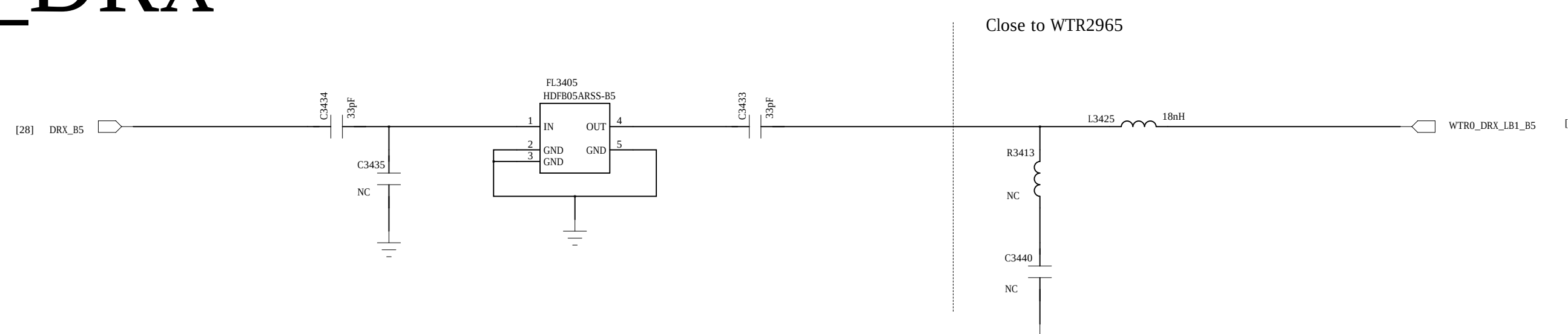
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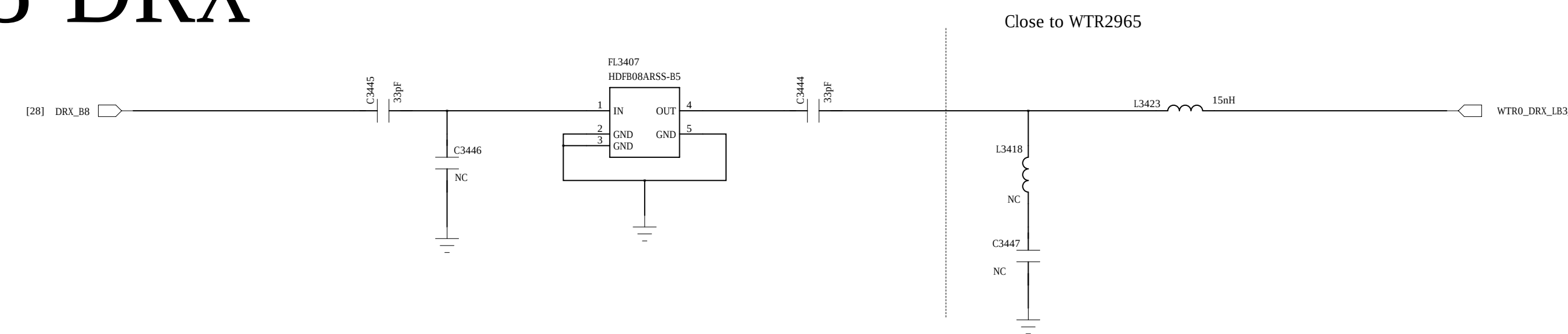
|B3_DRX



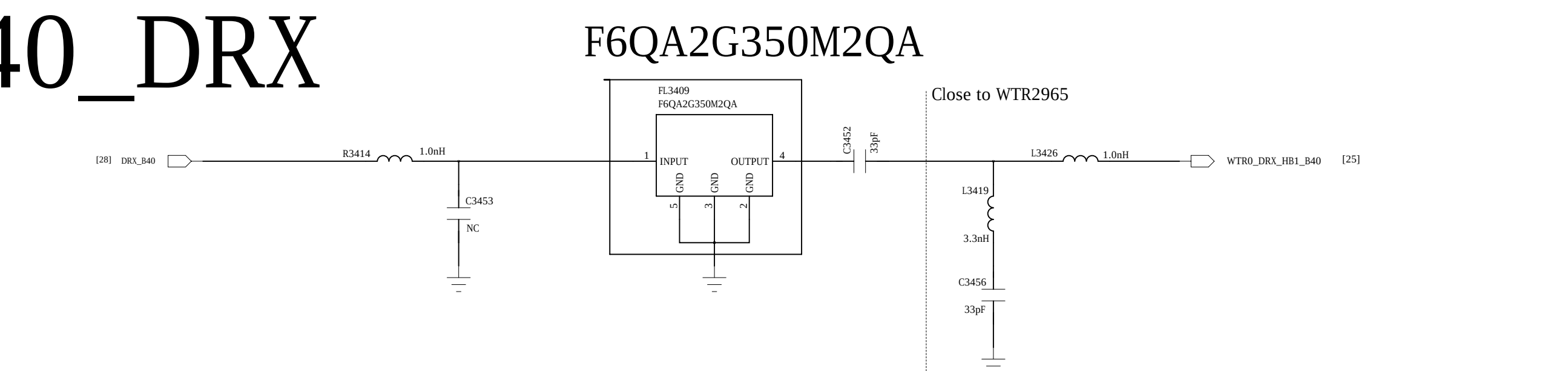
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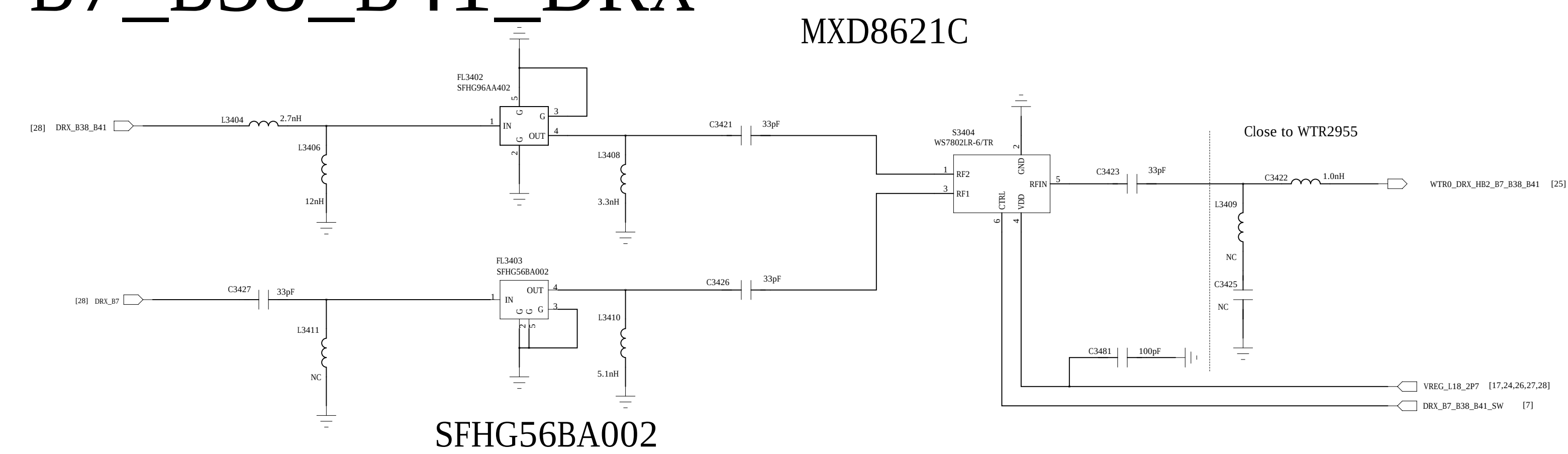
B8 DRX



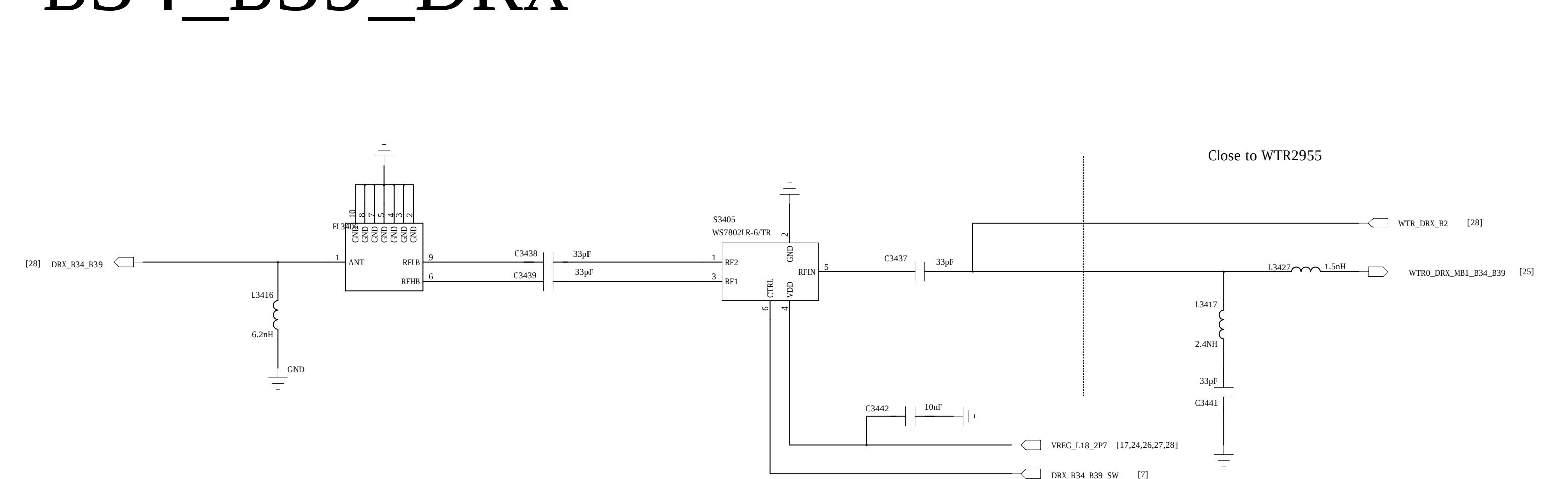
B40_DRX



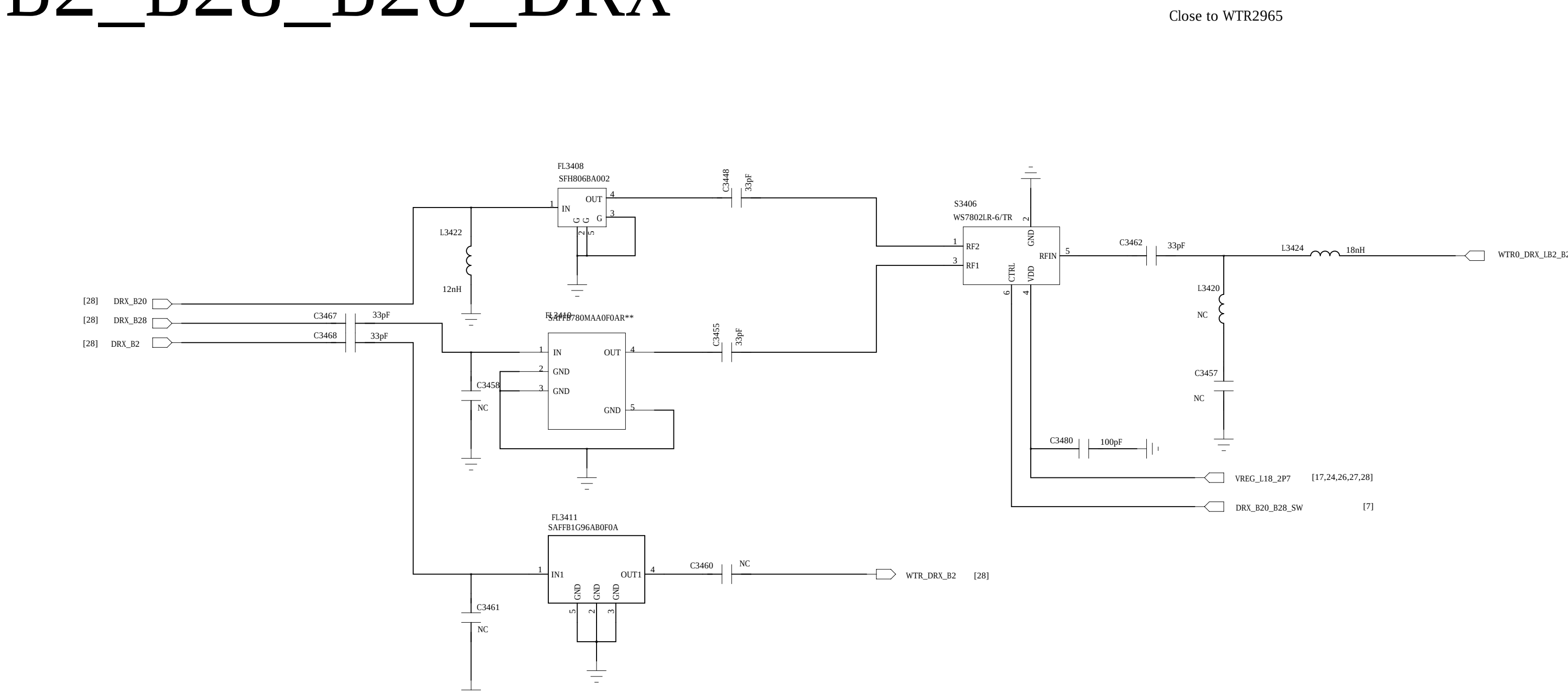
B7_B38_B41_DRX



B34_B39_DRX



B2_B28_B20_DRX



6

5

4

673

2

1

D

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C

C

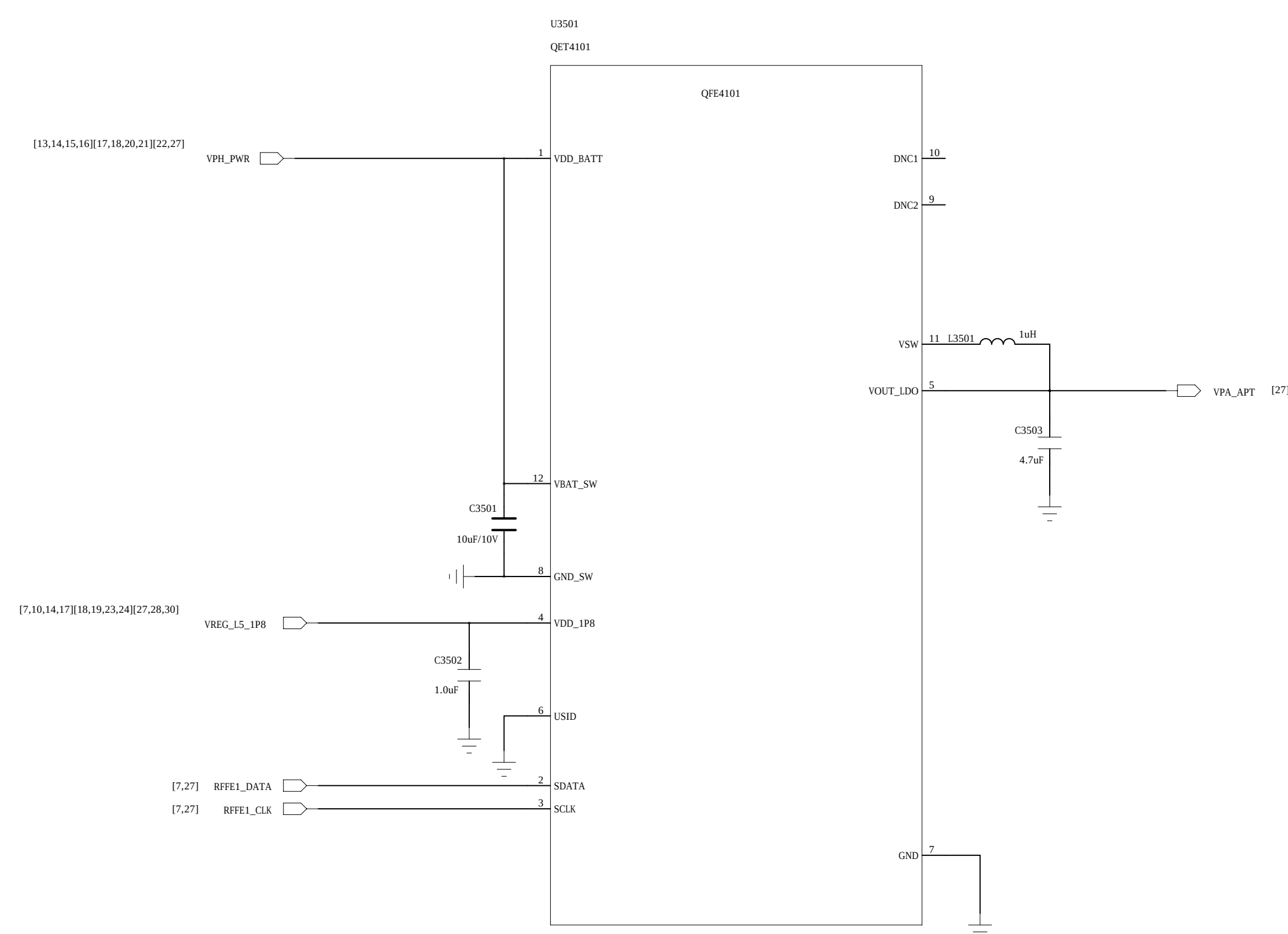
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B

A

A

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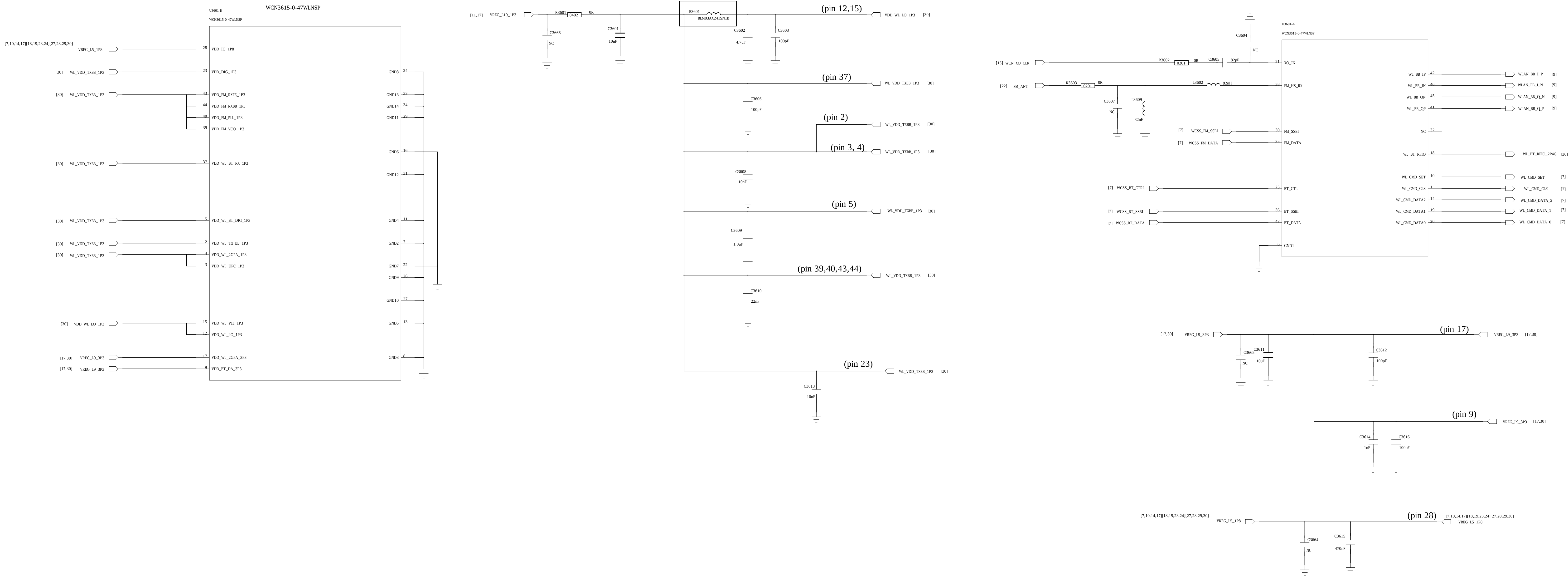
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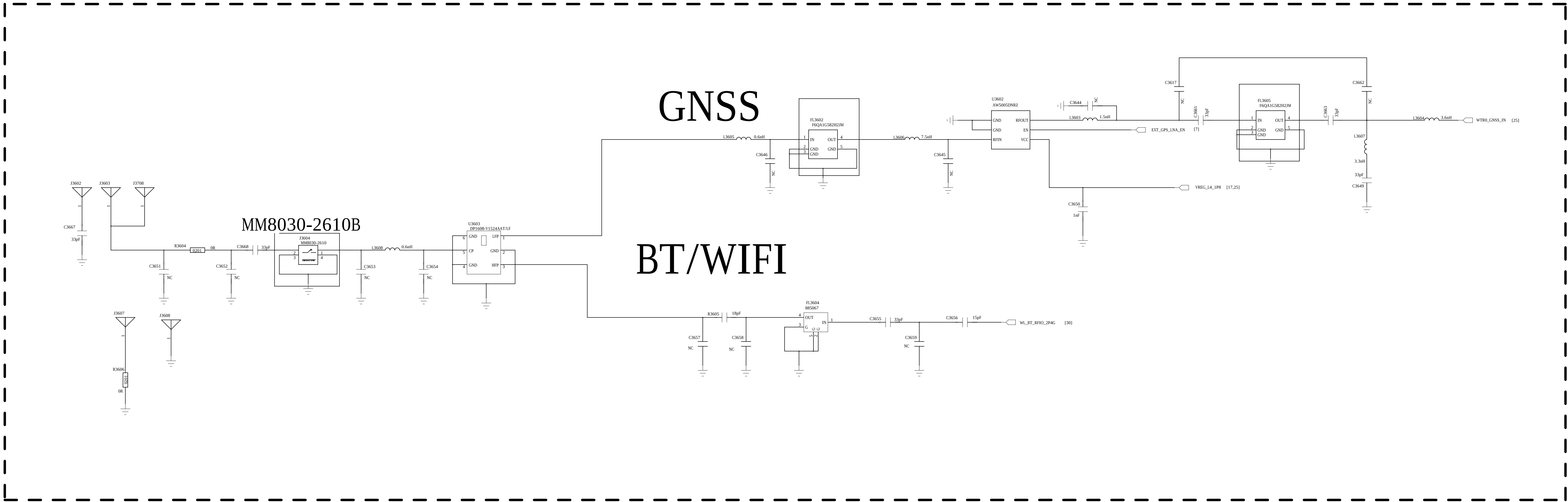
WCN

BLM03AX241SN1B



GNSS

BT/WIFI



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